

Using concentration measures for optimal screening of horizontal mergers^{*†}

Charles Taragin[‡]
US Department of Justice

Margaret Loudermilk[‡]
US Department of Justice

DRAFT: January 15, 2019

Abstract

Concentration measures like the Herfindahl-Hirschman Index are used as screening tools in horizontal mergers by antitrust enforcement agencies, but are these good measures of predicted competitive effects? This paper investigates the relationship between standard merger enforcement screening measures and predicted competitive effects important to economists such as industry prices. In doing so it advances understanding of the importance of model selection as well as the factors that drive variation in unilateral effects estimated by standard horizontal merger simulations. Finally, we consider optimal decision rules for merger enforcement based on economic concepts of cost and benefit.

Keywords: Herfindahl-Hirschman Index; upward pricing pressure; merger simulation; merger enforcement; unilateral effects

JEL classification: L13; L40; L41; K21

^{*}The views expressed herein are entirely those of the authors and should not be purported to reflect those of the US Department of Justice.

[†]We would like to thank Gloria Sheu, Eric Lewis, and seminar participants at the US Department of Justice Antitrust Division.

[‡]US Department of Justice, Antitrust Division, Economic Analysis Group, 450 5th St. NW, Washington DC 20530. Email: margaret.loudermilk@usdoj.gov.

1 Introduction

The Horizontal Merger Guidelines (HMG) provide a description of the principles that guide merger investigation and enforcement at the U.S. Department of Justice and Federal Trade Commission, the United States’ antitrust enforcement agencies. Within the HMG, a number of measures are used as thresholds or proxies to be considered in agency investigations and enforcement. For example, the post-merger level and increase in the Herfindahl-Hirschman Index (HHI) is cited as a screening tool for potentially anticompetitive mergers with thresholds for those considered “presumptively anticompetitive” or that “often warrant scrutiny”.¹ According to the Guidelines:

“The purpose of these thresholds is not to provide a rigid screen to separate competitively benign mergers from anticompetitive ones....Rather, they provide one way to identify some mergers unlikely to raise competitive concerns and some others for which it is particularly important to examine whether other competitive factors confirm, reinforce, or counteract the potentially harmful effects of increased concentration.”

Implicit in this statement is that not every merger can be investigated in depth. However, it is unclear how well these metrics map to measures of competitive effects such as prices, price-cost margins, and welfare that are central to modern economic analysis.

The assumption of a casual relationship between concentration measures and industry prices arises from the academic literature on the structure-conduct-performance paradigm. Examples include Borenstein 1989, Manuszak and Moul 2009, Manuszak and Moul 2008, Pinkse, Slade, and Brett 2002, and Schmalensee 1989. However, the primary modern tool for assessing predicted unilateral competitive effects due to a horizontal merger is merger simulation.²

While the sensitivity of merger simulation to assumptions about demand and supply specification is well-known (e.g., Weinberg and Hosken 2013, Crooke et al. 2003, Bass, Frank 2008), the relationship between common screening metrics such as HHI or upward pricing pressure (UPP) and predictions from merger simulation or observed post-merger prices is less well studied. Werden and Froeb 1996 investigates, in part, whether the combined share of the merging firms or post-merger HHI are useful predictors of price and welfare effects. In a differentiated product, Nash-Bertrand setting with Logit demand, Werden and Froeb find

¹U.S. Department of Justice and Federal Trade Commission 2010, p.19

²A survey of simulation methods for merger analysis can be found in Werden and Froeb 2006.

in a simulation study that both measures are at best mediocre predictors of price and welfare effects, which both exhibit variation that increases with the change in market concentration. In a retrospective analysis, Hausman and Sidak 2007 find no relationship between HHI and consumer prices for wireless telephone service in the EU. Miller et al. 2017 demonstrate that UPP acts as an accurate screen with log-concave demand specifications in Monte Carlo experiments but understates price effects under more convex demand assumptions. Similarly, Cheung 2016 used a cross-section of data on airline market routes to compare hypothetical horizontal merger predictions between UPP and merger simulation. Cheung found that the correlation between the UPP and merger simulation results increased as the size of the merging firms and cost savings from the merger decreased.

This paper contributes to understanding the relationship between standard merger investigation screening measures and competitive effects important to economists such as price and welfare and examines the factors that drive observed variation in estimated unilateral competitive effects. In addition, we consider an optimal decision rule for merger enforcement based on commonly used measures of concentration. The paper proceeds as follows: Section 2 briefly summarizes the set of models, concentration measures, and data generating processes employed by the simulation study developed to assess the extent concentration measures act as reasonable proxies for competitive effects in horizontal mergers. Section 3 reports the results of these numerical simulations. Section 4 considers an optimal screening rule for initiating horizontal merger investigations using concentration as a proxy for anticompetitive harm, and section 5 concludes.

2 Monte Carlo Experiments

2.1 Demand and Supply Specifications

Here, we describe the game-theoretic models that underpin our numerical experiments. These include: Cournot with quadratic costs and log-linear as well as linear demand; Bertrand with Almost Ideal Demand (AIDS), Constant Elasticity of Substitution (CES) demand, and Logit demand; and a second score auction with Logit demand.³ We selected the Cournot, Bertrand and 2nd score auction models for two reasons. First, versions of these models have been employed by both the FTC and the DOJ in publicly litigated matters. For example, in

³See Appendix B for details of the demand and supply model specifications.

2011 the Division used a Bertrand pricing game with linear demand to simulate the effects of a merger between HR Block and TaxAct, two firms that specialize in tax preparation. In 2017, The Division also employed a Bertrand model, this time to simulate the effects of a horizontal merger between Aetna and Humana, two health insurers. In that same year, the Division used a 2nd score auction model to simulate the effects of a merger between Anthem and Cigna, two other large health insurers. Finally, in 2018, the FTC used a Cournot model to simulate the effects of a merger between Tronox and Cristal, two of the largest producers of titanium dioxide. Second, while other papers have established that for the Bertrand model, the curvature of the demand curve can deeply affect the magnitude of the post-merger price changes, to our knowledge, little research has been done analyzing the effect that the assumed form of the strategic game has on post-merger price changes.

For Bertrand, we selected the AIDS, Logit and CES demand systems for three reasons. First, like log-linear demand in Cournot, AIDS can have large second derivatives, particularly at high prices. In contrast, the Logit model and to a lesser extent the CES model generally have smaller second derivatives. Including these three specifications allows us to assess the effect of demand curvature for a given model of supply. Second, while the choice of demand specification is heavily influenced by the particulars of an industry, data limitations often influence the set of models that can be implemented in practice, and the data requirements of these models differ. Typically, both AIDS and CES demand are estimated using expenditure shares while Logit demand is estimated using quantity shares. Third, for all three of these demand systems we assume that at least in the pre-merger state of the world, consumer substitution patterns are a function of either revenue or quantity shares, a feature which is easy to incorporate into these three demand systems.

We selected the linear and log-linear demand systems for Cournot because in our experience, the Cournot model is often used to simulate merger effects in homogeneous good industries, and the linear and log-linear demand systems capture this feature nicely while also allowing us to explore two extreme cases with regard to curvature. Linear demand has no curvature (second derivatives equal 0), whereas log-linear demand exhibits substantial curvature, particularly at high prices. Logit demand is used in the 2nd score auction model largely due for convenience and tractability but is also one of the few functional forms previously used in published work.

Also of interest are the demand models that we did not include, most notably log-linear and linear demand for Bertrand. Without additional restrictions these demand systems require estimation of a large number of parameters, and it is often the case in practice that there

is insufficient data to obtain reliable estimates. Further, unlike AIDS, Logit, and CES, without further restrictions log-linear and linear demand are not necessarily consistent with the axioms of consumer choice theory. As a result, it is difficult to calculate welfare measures that are important in assessing the competitive effects of a merger.

2.2 Measures of Competitive Harm

A number of indicia are commonly used to proxy for the anticipated competitive effects of a horizontal merger, including industry firm counts, merging party size rank, combined party market share, HHI, UPP, and compensating marginal cost reduction (CMCR). Many of these indicia consist of two parts: an algorithm for calculating the index and a threshold decision rule for determining whether a horizontal merger is likely to raise competitive concerns. Here we describe the “thought experiment” underlying each of these indicia, as well as some of their properties, advantages and drawbacks. Before doing so, it is important to bear in mind that by construction these indicia only quantify the price effects due to unilateral incentives driven by diversion between the merging parties. Incorporating the effects of other concerns, such as issues related to dynamics (product repositioning, entry), coordinated effects, vertical concerns, etc. requires either modifications or entirely different indicia. Such extensions are beyond the scope of this paper.

2.2.1 Firm Count

The intuition underlying the “firm count” indicia is that a merger reduces the number of firms in a market from N to $N - 1$, which would tend to raise prices and harm consumers, *ceteris paribus*. The magnitude of the expected price effect is expected be small when N is large and large when N is small. Calculating this indicia is easy: 1) Identify firms in market; 2) Count the firms (N); and 3) Reduce the number of firms by 1 ($N - 1$).

Interpreting the Firm Count measure is more difficult. On one hand, it seems plausible that a merger reducing the number of firms from 100 to 99 is unlikely to yield substantial harm. On the other hand, it is plausible to believe that a merger reducing the number of firms from 2 to 1 yields substantial harm. But what about a 6 to 5 merger? A 4 to 3 merger? In addition, this measure assumes each firm included in the market is as competitively significant as any other firm (i.e., Firms are in some sense “symmetric”)., but what if some firms have more sales than others? What if manufacturing costs vary across firms, consumers value the

attributes of each product differently, or some firms offer a portfolio of products while others only sell a single product? It seems plausible that most if not all of these factors distinguish one firm from another in ways that render this measure noisy. The firm count measure also provides no guidance as to which firms should be included in the antitrust market of interest. Rather, it assumes that the contours of the market are clearly identified, which is rarely true in differentiated product markets.⁴

2.2.2 Merging Party Size Rank

The “Party Rank” measure, which identifies the size rank of the merging parties (e.g. the merging parties are the 1st and 3rd largest firms in the market), addresses one criticism of the firm count metric by (partially) internalizing the potentially asymmetric nature of the merging parties. A merger reducing the number of asymmetric firms in a market from N to $N - 1$ may raise prices by a substantial amount, depending upon the sizes of the merging firms. The magnitude of the price effect will be small when either both merging parties are small and therefore have low rank OR one party has a high rank and the other a low rank. Thus, Party Rank attempts to capture the difference between the the 1st and 2nd largest firms merging out of ten from the 9th and 10th largest merging. In this example, the Firm Count for both mergers is 10 to 9 although we would expect larger competitive effects from a merger of the market’s largest firms.

2.2.3 Merging Party Combined Share

One important drawback of both the Firm Count and Party Rank measures is that they do not capture information about the magnitude of the size differences between firms. The “Party Share” measure, which equals the combined market share of the merging parties, remedies this by explicitly using information on the size of the merging parties. The intuition behind this measure is straightforward: all else equal, mergers where the parties have a higher combined share control more of the market post-merger and therefore have a greater incentive to raise prices.

Aside from its intuitive appeal, one useful feature of the party share measure is that the US Courts as well as US antitrust enforcement agencies have at times recognized the importance of this indicia. In *Philadelphia National Bank*, the US supreme Court ruled that mergers

⁴This criticism of Firm Count is true of nearly all measures of competitive harm we consider and thus is not repeated in each of the subsequent sections.

where the combined party share is greater than 30% are presumptively anti-competitive. Moreover, the now defunct 1992 Horizontal Merger Guidelines published by the US FTC and DOJ indicate that if the combined party share is greater than 35%, then the agencies will view the merger as presumptively anti-competitive. While the 2010 HMG no longer enumerates a combined share threshold, the HMG still recognizes the import of party shares, stating “The Agencies also may consider the combined market share of the merging firms as an indicator of the extent to which others in the market may not be able readily to replace competition between the merging firms that is lost through the merger.”⁵

Although intuitive, the Party Share measure has two drawbacks. First, it does not distinguish between different market compositions that could affect competitive interactions such as between markets with a few large non-merging parties and many versus a market with only many small non-merging firms. Second, this measure does not provide a way to quantify the magnitude of harm associated with a particular concentration level.

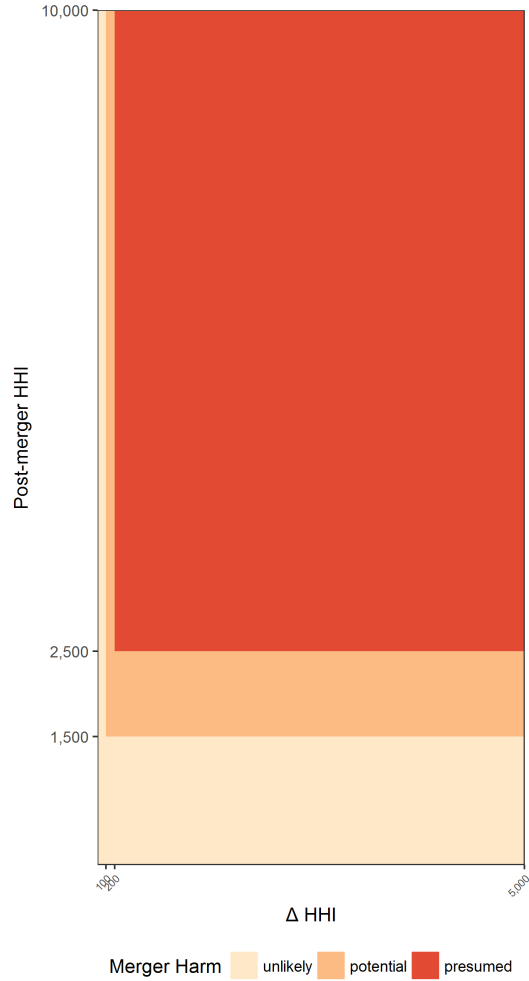
2.2.4 Herfindahl-Hirschman Index

The Herfindahl-Hirschman Index not only remedies the critics of the Firm Count and Party Rank measures by explicitly using information on firm size but also quantifies the differences in market composition that remains a weakness of Party Share. Specifically, the HHI is calculated as the sum of the squared shares of firms in the relevant market, yielding an indicia that is between 0 (perfect competition) and 10,000 (monopoly). It can be shown algebraically that the post-merger change in the HHI, ΔHHI , is given by 2 times the product of the merging firms’ market shares.

The HHI has a number of useful theoretical properties. First, it may be derived from a standard Cournot quantity-setting game, one of work-horse game-theoretic models in the IO literature. Second, the HMG articulate criteria describing under what conditions a merger may or may not be cause for concern in terms of the HHI. As shown in Figure 1 if the post-merger $HHI \geq 2,500$ and $\Delta HHI \geq 200$ then, according to the HMG, the merger should be considered *presumptively* anticompetitive. If the post-merger $HHI \geq 2,500$ and $100 < \Delta HHI \leq 200$ OR post-merger $2,500 > HHI \geq 1,500$ and $\Delta HHI > 100$ the merger may be *potentially significantly* anticompetitive. If the post-merger $HHI < 1,500$ then it is considered *unlikely* to have adverse competitive effects. A $\Delta HHI = 200$ could result, for example, from the merger of two firms each with a 10% market share.

⁵U.S. Department of Justice and Federal Trade Commission 2010, p.???

Figure 1 HMG HHI-based decision rule



Another feature of the HHI is that it can be readily converted into a measure of the effective number of symmetric firms in a given market, $N_{eff} = 10,000/HHI$. Among other things, this useful result aids in interpreting the HMG's HHI thresholds since a post-merger HHI of 2,500 implies 4 equal-sized firms, while a 1,500 HHI implies roughly 7 equal-sized firms. Therefore, loosely speaking, the HMG state that mergers resulting in 4 or fewer equal-sized firms in a market are presumptively anticompetitive, while mergers where 7 or more firms remain in the market are unlikely to be anticompetitive. Despite these useful properties, the HHI remains silent on a few key issues such as how much harm is generated at the Guidelines' thresholds, and what is the right measure of output (revenues or quantities) for calculating market shares.

2.2.5 Upward Pricing Pressure

UPP attempts to directly quantify the competitive effects of a horizontal merger and is derived as a first order approximation of the merging parties equilibrium pricing strategy. Intuitively, the UPP measure asks ‘When a firm raises its price on a product, how much of the sales it loses will shift to its merging partner (diversion), and how much are those sales by the merging partner worth (margin)?’ Recapturing sales diverted from the merging partner lessens the cost of raising price, creating “upward pricing pressure”.

UPP is often quantified by the Generalized Upward Pricing Pressure Index (GUPPI). The GUPPI is obtained by expressing the value of a firm’s sales diverted to the merging partner as a fraction of the firm’s price. Calculation of the GUPPI requires the prices of both firms, one of the firm’s price-cost margins, and a measure of diversion between the merging firms under the assumption that firms are competing through price setting (Nash-Bertrand Competition).

In terms of linking to unilateral effects resulting from a horizontal merger, there is no direct meaning in a particular value of the GUPPI, except a larger number indicates a larger incentive to raise price post-merger. This has the benefit of allowing comparison to a common benchmark and some practitioners have suggested a “safe-harbor” level of 5-10%. However, each merger can result in a multiplicity of GUPPI figures. Two merging firms with a single, homogeneous product in a single market result in two UPP numbers. If each firm has multiple products, two numbers are created for each product in each market. There is no theoretical or empirical guidance regarding which of the resulting numbers to use in evaluating the competitive effects or on how to weigh different results across products or markets.

2.2.6 Compensating Marginal Cost Reduction

The compensating marginal cost reduction (CMCR) considers the dual problem to those discussed previously. Instead of asking by how much prices will rise after a merger, the CMCR can be thought of as the level of cost efficiencies needed to cancel out the upward pricing pressure incentive. CMCR uses the same inputs as the GUPPI and like UPP can produce multiple values. However, the CMCR is an exact prediction of the unilateral effects of a horizontal merger - not a first approximation like UPP. A further benefit of CMCR is that the formula can be modified for use under Cournot competition. For two symmetric single-product firms with the same price-cost margins (m), prices, and diversions (d), CMCR

reduces to

$$CMCR = \frac{m}{1-m} \left(\frac{d}{1-d} \right)$$

2.3 Data Generation

In order to assess the efficacy of the measures in Section 2.2, we use the models described in Section 2.1 to simulate the effects of a large number of horizontal mergers across different market conditions. For all simulations, we assume that in the pre-merger state there are $N \in \{3, 4, 5, 6, 7\}$ single-product firms “inside” the market of interest and an outside good. The firms and the outside good are assumed to interact strategically in the pre-merger state, but only the inside goods change their prices in the post-merger state. The single-product firm shares are drawn from a Dirichlet distribution with all N concentration parameters equal to 2.5. Setting the concentration parameters in this fashion yields firm shares with mean $1/N$ and variance equal to $\frac{N-1}{2.5N+1}$. In addition, we assume that in every simulated market the outside firm’s product costs \$6 and that the outside firm earns a margin of \$2. Finally, the outside firm’s share is drawn from a Uniform Distribution restricted to be between 0.1 and 0.7.⁶ For Cournot, these assumptions are sufficient to identify all demand and cost parameters. For, Bertrand and the second score auction, however, these assumptions are not sufficient to separately identify firm marginal costs and prices. To remedy this, we assume for the Bertrand and Auction models that marginal costs are equal to \$2 for all firms⁷.

We simulate 50,000 markets for each demand-supply model specification described in Section 2.1. For each market, a horizontal merger is created by randomly assigning two firms as the merging parties. Among these two firms, the one with the largest market share is designated as the acquirer and the other as the target firm. In all our models, a horizontal merger is represented as an ownership change, with the merging parties’ products (or plants) placed

⁶We chose this range in order to determine how sensitive merger effects are to the attractiveness of the outside good. We limit the range of outside shares to $[0.1, 0.7]$ because our parameter calibration strategies rely heavily on the value of the outside share, and outside shares beyond this range tended to yield more markets with implausible model parameters (i.e. negative marginal costs for Cournot) or implausibly large price effects.

⁷Doing so guarantees that all firm prices and marginal costs are positive and that firms’ pre-merger prices are not too close to zero, thereby preventing price changes expressed as a percentage of pre-merger prices from becoming explosively large.

under common control.

Next we test that each simulated market meets the criteria to be both a valid antitrust market, using the Hypothetical Monopolist Test (HMT) at the 5% level, and to require pre-merger notification under the Hart-Scott-Rodino Act, referred to subsequently as an HSR filing. In the context of our numerical simulations, the HMT flags markets that do not contain all the alternatives that consumers would plausibly view as substitutes to products included in the market. This test is important because all of the indicia discussed here are calculated using data from firms included in the market, and if the market does not contain all plausible alternatives, the indicia are more likely to incorrectly identify a merger as anti-competitive when it is not.

An HSR filing is required under the Hart-Scott-Rodino Act in 2018 if at least one party has annual revenues of at least 168.8 million USD and the other has revenues of at least 16.9 million USD.⁸ To implement this restriction we assign the acquiring firm the HSR threshold of 168.8 million USD. Doing so implies both a total market size and the revenues of the target firm. If the revenues of the target firm are greater than or equal to 16.9 million USD, we assume that the merger meets the HSR filing requirements.

Eliminating markets that do not meet both these criteria results in approximately 222,000 simulated HSR reportable, antitrust markets. Henceforth, this set is referred to as the ‘sample markets’. Restricting our sample in this way allows us to focus on the performance of indicia in the set of markets that would be relevant to antitrust enforcement authorities. Table 1 summarizes the competitive effects implied from such horizontal mergers across all the sample markets.

Table 2 summarizes the salient features of the sample markets at the 5th, 25th, 50th, 75th, and 95th percentiles. Market revenues are reported in millions of US dollars, and the outside share is reported as a percentage. This table reveals that our numerical simulations represent a diverse set of markets, with annual market revenues typically between \$300 million and \$2.8 billion, the distribution of outside shares appearing roughly uniformly distributed between 0.1 and 0.7, and market elasticities ranging from very elastic (-6) to very inelastic (-0.4).

Table 3 presents statistics for the market indicia calculated from the sample markets. The range of values for Firm Count, Party rank, Party Share, and HHI arise largely by construction from the data generating process described above. The distribution of these values for

⁸There are additional requirements for “size of transaction” that consider the value of voting securities and assets of the merging parties. These considerations are outside the scope of the paper. See ???.

Table 1: Simulated markets outcomes, by model

Outcome	Supply	Demand	5%	25%	50%	75%	95%
Consumer Harm (\$)	Cournot	Log	8.2	23	54	144	631
		Linear	4.4	13	31	73	233
	Bertrand	Aids	1.4	4.4	10	23	87
		Logit	2.3	5.6	10	18	37
		Ces	0.54	1.7	4.1	9.2	24
	Auction	Logit	1.6	3.8	6.8	12	28
Industry Price Change (%)	Cournot	Log	0.59	1.8	4.3	11	42
		Linear	0.29	0.99	2.3	5.1	13
	Bertrand	Aids	0.12	0.55	1.6	4.4	21
		Logit	0.2	0.69	1.6	3.4	9.2
		Ces	0.13	0.45	1.0	2.4	7.3
	Auction	Logit	0.14	0.47	1.1	2.4	7.5
Merging Party Price Change (%)	Cournot	Log	0.59	1.8	4.3	11	42
		Linear	0.29	0.99	2.3	5.1	13
	Bertrand	Aids	0.42	1.3	3	7	28
		Logit	0.96	2.2	3.8	6.4	13
		Ces	0.59	1.4	2.6	4.8	11
	Auction	Logit	0.71	1.5	2.6	4.6	10
Producer Benefit (\$)	Cournot	Log	6	15	34	87	373
		Linear	3.3	9	21	48	146
	Bertrand	Aids	0.47	1.7	5	14	49
		Logit	0.74	2	4.8	10	23
		Ces	0.38	1	2.4	5.1	11
	Auction	Logit	1.6	3.8	6.8	12	28

the 5th, 25th, 50th, 75th, and 95th percentiles are reported along with those for ΔHHI , UPP, and CMCR.

The correlation matrix for these market indicia is shown in Table 4. There is significant positive correlation between all pairs of measures, which is unsurprising due to the algebraic relationships between them. Yet, the correlations provide some quantification of the potential for combinations of indicia to provide additional information in the merger screening process. For example, if the merging parties' combined share is known then little additional information is gained by calculating the change in HHI resulting from the merger (correlation is 0.96 between the measures), but also knowing the upward pricing could add valuable information if UPP does a good job of screening for unilateral competitive effects (correlation is 0.78). Similarly, if the only measure known is Firm Count then any indicia other than HHI could provide significant additional information.

Table 2: market summary

Feature	5%	25%	50%	75%	95%
Market revenues	319	501	743	1,176	2,753
Outside share	13	24	37	52	66
Market elasticity	-6.1	-2.3	-1.4	-0.81	-0.39

Table 3: Indicia summary

Index	5%	25%	50%	75%	95%
Firm Count	3	4	5	6	7
Party Gap	1	1	2	3	4
Party Share	18	30	41	56	79
HHI	1,996	2,523	3,306	4,689	6,701
HHI Change	117	349	697	1,334	2,774
UPP	1.2	3.4	6.3	11	24
CMCR	1.6	5.7	13	30	118

Table 4: Correlations among indicia (Spearman)

	Party Gap	Party Share	HHI	HHI Change	UPP	CMCR
Firm Count	0.37	0.69	0.90	0.65	0.56	0.57
Party Gap		0.21	0.39	0.31	0.32	0.29
Party Share			0.85	0.96	0.78	0.80
HHI				0.81	0.68	0.70
HHI Change					0.82	0.82
UPP						0.99

3 Results

The results from our numerical experiments are displayed in Figures 2 - 10. Our aim with these plots is to assess how effective each of the aforementioned indicia are at predicting the effects of a horizontal merger as compared to merger simulation. Except for Figures 3 and 8, all figures are violin plots that depict the distribution of simulated industry price changes as a function of the discretized values of a particular indicia for the 6 different model specifications. We chose violin plots because they do an excellent job at displaying asymmetric distributions. For a particular discretized value, the horizontal lines in each violin plot depict the 25th, 50th, and 75th percentile industry price changes. To make these plots more readable, we also chose to censor the industry price changes at the 5th and 95th percentile across all model specifications.

3.1 Firm Count

Figure 2 depicts the distribution of industry price changes by Firm Count. First and foremost, the plot reveals that across all models, markets with fewer firms have exponentially larger average effects on industry-wide prices than markets with more firms. However, markets with fewer firms also have a larger range of plausible price effects than markets with a larger number of firms. Thus, while the distribution of price effects is positively skewed for all values (i.e., skewed toward larger industry price effects), the distributions are most skewed in markets with fewer firms. This result is intuitive since firms have the greatest potential to exercise market power in concentrated markets. Relatedly, holding the number of firms constant, skewness appears to decrease as one moves from the right-most model (Cournot, log-linear demand) to the left-most model (Auction, Logit demand). We conjecture that the reason for this skew is that the market elasticity, which is not well captured by this index, plays a more prominent role in predicting merger effects under Cournot than under the Bertrand or Auction models.

Notice that at the 5 to 4 level (highlighted in orange), representing the implied current threshold, all model specifications produce mean industry price changes of less than 5%, and more than 95% of all the simulations have industry-wide price effects less than 10%. In contrast, the likelihood of a negligible industry price effect (less than 1%) is very small for a 3 to 2 merger. Additionally, the probability of a negligible effect increases with the number of pre-merger firms in the market as illustrated by the increasing mass below 1% in each

plot as Firm Count increases in each panel.

Taken together these observations suggest that a significant amount of information about potential price effects is conveyed by the Firm Count measure. Both the likelihood of a significant price effect and the predicted size of the effect increase as the number of pre-merger firms in the market decreases. However, the measure also becomes increasingly noisy as the number of firms decreases. This is consistent with the observations in Section 2.2.1 of factors that could substantially vary competitive conditions between markets with the same number of firms and as the number of firms.

3.2 Party Rank

Figure 3 displays a heat map summarizing the relationship between each of the merging parties' size rank, as measured by market share, and median industry price changes, with darker blue shading corresponding to larger price effects. These plots reveal that that under the models considered the largest price effects typically occur when firms are closely ranked. For example, in the Cournot model with log-linear demand, the largest median price effects occur when the merging parties are ranked 2nd and 3rd, while some of the smallest price effects occur when one party is ranked 1st and the other is ranked 7th in terms of size. Notice, however, that while markets where the merging parties are ranked sequentially typically have the largest price effects, the magnitude of the price effects tend to decrease as the rankings of both parties increase. For example, the median price effects are typically larger when the parties are ranked 2nd and 3rd than when they are ranked 4th and 5th. These results are intuitive for largely algebraic reasons since the greatest gain in combined market share, and thus concentration, occurs when firms are similar in size. Finally, notice that the rate at which these merger effects decay appears to vary across models, with the right-most model (Cournot,log-linear) exhibiting the slowest rate of decay, and the left-most model (Auction, Logit) suggesting the fastest rate of decay.

To further quantify the relationship between Party Rank and competitive effects, we create a measure, denoted Party Gap, that attempts to captures the degree “symmetry” between the merging parties. We define Party Gap as the difference in the Party Rank of the merging parties so that a merger between the 2nd and 3rd largest firms by market share has a Party Gap of 1. Figure 4 presents Party Gap versus changes in industry-wide prices, revealing a now familiar pattern. As the merging parties grow closer in rank both the average price effects and the variance of the predicted price effects increase. However, in expectation

only mergers with a Party Gap equal to 1 are likely to produce industry price effects of more than 5%, which suggests that this measure is of limited use in screening potentially anticompetitive mergers. This is unsurprising since the measure can not differentiate between a merger between the 1st and 2nd largest firms in the industry and the 6th and 7th largest and as a result incorporates only part of the information usually conveyed by the qualitative Party Rank measure.

3.3 Party Share

Figure 5 displays the relationship between the pre-merger combined share of the merging parties and industry price changes. Highlighted in orange is the 30% threshold, which has been historically identified as the threshold for establishing that a merger is presumptively anti-competitive. Similar to the “Firm Count” measure, we see that while median price effects from the simulated mergers increase at an exponential rate as party combined shares increase, so does the plausible range of price effects. Indeed, even the model with the smallest range of effects (Bertrand, CES) has industry-wide price effects ranging roughly between 5% and 15% when the parties combined share is 90%.

Finally, at the 30% threshold, notice that across all the models, approximately 95% or more of the simulated markets experience less than a 5% price effect, and half of all markets experience less than a 2% price effect. This suggests that, if used alone, this threshold would lead to investigating a large number of mergers with very small predicted price effects as compared to merger simulation results.

3.4 HHI

Figure 6 presents the distribution of estimated industry price changes by the level of the post-merger HHI for each model. Across all models, both the level and the variance of the quantiles of the estimated industry price changes increase with the change post-merger HHI. However, it is apparent that the choice of model matters with the greatest variance in the effects exhibited by the model with Cournot supply and log demand. In contrast, the Logit demand model - the workhorse model for merger simulation - exhibits the significantly less variation in price effects in both the Bertrand and auction supply settings. Figure 6 also illustrates that all models produce significantly larger price effects in a small fraction of the simulated markets with price effects at the 95th percentile generally more than double the

median value.

The results are nearly identical in Figure 7, which presents the distribution of estimated industry price changes by the change in the post-merger HHI. Although the change in HHI has an additional benefit. The change in the post-merger HHI is associated with a stronger negative correlation with a negligible price effect. This makes the change in HHI more useful as a screening metric since it is likely to lead to fewer investigations of mergers that would not prompt an enforcement action.

Together Figures 6 and 7 suggest that there is little additional information about projected price effects to be gained by using the combination of the post-merger HHI level and the change. However, the HMG categorizes mergers by the combination of the level and change in industry concentration. Figure 8 investigates whether there is an informative relationship between pre-merger HHI vs. post-merger change in HHI across models as illustrated by a heatmap. While Figure 8 shows price effects are more likely as both measures increase significant effects are only predicted on average at very high values. Thus, there is no relationship between the two measures to clearly inform merger enforcement decisions.

3.5 UPP

Figure 9 displays the relationship between UPP and industry-wide price changes across the different models. First, these figures confirm previous findings that UPP’s predictive power substantially degrades as demand curvature increases. The predictions exhibiting the least variation in price effects are given by the Cournot-linear and Bertrand-Logit specifications, which have little or no curvature, and the greatest variation for Cournot-log and Bertrand-Aids specification, which have substantial curvature. Second, with the exception of Cournot-log and Bertrand-AIDS specifications, UPP tends to over-predict the price effects from a merger. For example, even in the Cournot model with linear demand, where UPP performs generally well, industry price changes in the 10-20 bucket (labelled “10” on the abscissa) are almost never above 15% and for a number of models (i.e Bertrand, Logit) are almost never above 10%. This appears to generally support the suggestion of a “safe-harbor” value or other similar screening threshold based on UPP of approximately 5-10%. In addition, we observe that the mean and variance of the predicted price effect increase with the degree of upward pricing pressure. We posit that this is because the potential importance of 2nd order effects, not captured by UPP, grow as price effects increase.

3.6 Bertrand CMCR

Figure 10 displays the relationship between the Bertrand CMCR and industry-wide price changes across the different models. Before discussing the results, it is important to reiterate that CMCR is designed to predict the amount by which merging parties' costs would have to fall in order to offset *any* price increase, and is therefore not explicitly designed to be used to predict merger price effects. However, as with the other indicia, CMCR does have some predictive power. As the measure increases, the distribution of industry price changes increases as well.

Unfortunately, as with many of the other indicia discussed above, the range of price effects also increases as the CMCR increases, with models having substantial demand curvature (i.e. Cournot, with Log-linear demand and Bertrand with AIDS demand) experiencing the largest increases in price dispersion. Also, like UPP, CMCR tends to under-predict industry-wide price effects. For example, even in the Bertrand model with Logit demand, where one might expect the CMCR to perform well, industry price changes in the highest reported bin are never above 10%.

4 Optimal screening rules for investigating horizontal mergers

In this section, we develop a series of “optimal” decision rules for determining whether to investigate a horizontal merger based on the concentration measures previously analyzed. It is important to emphasize that all of the rules discussed hinge on two important assumptions: 1) the data generating process underlying the population of horizontal mergers considered, and 2) the behavioral model for weighing the costs and benefits of a particular decision rule. Initially, we will assume that the population of potential mergers consists of the markets in our numerical experiments. In other words, we will assume that all horizontal mergers can be described by the data generating process used to generate our numerical experiments. Later, we will relax this assumption and consider how changing the composition of models affects the decision rules. It is important to stress that the behavioral models we consider here are relatively simple heuristics designed to weigh the harm from not pursuing a potentially anti-competitive merger against the benefit from not investigating relatively benign mergers with both measures construed very narrowly. None of the behavioral models analyzed here take

into account the wealth of other considerations that antitrust enforcement agencies might also incorporate.

4.1 Type I/Type II error analysis

Suppose that the agencies are interested in investigating horizontal mergers that yield substantial harm to consumers but are also interested in *not* investigating horizontal mergers that are unlikely to yield substantial harm. Further, suppose that price thresholds for determining what constitutes substantial harm and what does not are well defined.

Formally, Let V_H denote the average benefit an agency receives from investigating a harmful merger and V_S denote the average cost an agency incurs from investigating an innocuous merger. Further, suppose that a merger is considered harmful if it is expected to increase average industry prices $\Delta\bar{p}$ by at least Δp_H percent, while a merger is considered innocuous if it expected to increase $\Delta\bar{p}$ by no more than Δp_S . Then, for a given index I agencies choose threshold D_I to maximize *ex ante* net benefit from investigating future mergers:

$$\max_{D_I} V_H Pr(\Delta\bar{p} \geq \Delta p_H | D_I) - V_S Pr(\Delta\bar{p} < \Delta p_S | D_I).$$

For continuous decision rules, taking the derivative of the above w.r.t to D_I and re-arranging yields:

$$\frac{V_H}{V_S} = \frac{\frac{dPr(\Delta\bar{p} < \Delta p_S | D_I)}{dD_I}}{\frac{dPr(\Delta\bar{p} \geq \Delta p_H | D_I)}{dD_I}}, \quad (1)$$

$\frac{V_H}{V_S}$ represents how enforcement agencies value preventing anti-competitive mergers relative to not investigating benign mergers, which we will refer to as the “enforcement ratio”. Here we explore two cases. First, we consider the case when $\frac{V_H}{V_S} = 1$. In this case, agencies place equal weight on these two outcomes and consequently the thresholds are only a function of the relative incidence of harmful and benign mergers. Second, we consider the case when $\frac{V_H}{V_S} = 10$, which is the value at which the Party Share and Post-merger HHI decision thresholds predicted by this model closely match observed Party Share and Post-merger HHI decision rules. This value of the enforcement ratio would suggest that agencies put much

greater weight on preventing potentially harmful mergers than on investigating benign ones.

In determining the optimal screening thresholds, we assume that any horizontal merger that yields at least a 5% industry-wide price increase is anti-competitive, while any horizontal merger yielding no more than a 1% industry-wide price increase is benign.⁹ Figure 11 displays the cumulative proportion of simulated markets that for a given value of an index are i) above the 5% threshold (solid dark blue) and ii) below the 1% threshold (solid light blue). Note that with the exception of the Firm Count measure, the cumulative distributions are reflected so that more markets satisfying the thresholds are present for lower values of a particular index. These plots reveal that roughly 20% of the simulated markets have more than a 5% industry-wide price effect, while almost 40% of the simulated markets have less than a 1% price effect.

The green (dot-dashed) vertical line displays the most current threshold for each index (if available). For example, the 30% “Party Share” rule occurs about where the 5% and 1% thresholds cross near the maximum of the 5% rule, indicating that under our modelling assumptions, the 30% rule ensures that all harmful mergers are investigated while about half of all benign mergers are also investigated. By contrast, the 200 point “HHI Rule” also investigates all the harmful mergers, but also investigates almost 70% of all benign mergers.

The pink (dashed) vertical line displays the optimal decision rule for each index under the assumption that $\frac{V_H}{V_S} = 10$, so the agencies value preventing merger harm 10 times more than not investigating benign matters. As this relative valuation was obtained by matching the predicted thresholds to the observed thresholds for Party Share and Post-merger HHI, it is perhaps unsurprising that those thresholds match so closely. Interestingly, however, the Firm Count threshold is predicted to be 6, which is greater than the threshold of 5 that is inferred from the HHI level with symmetric firms. Likewise, the HHI Change threshold is greater than 400, more than twice the magnitude of the current 200 threshold. Also of interest are the UPP and CMCR thresholds, which are about 5% and 9%, respectively, and in line with proposed “safe-harbor” values in the literature. The red (dotted) vertical line displays the optimal decision rule for each index under the assumption that $\frac{V_H}{V_S} = 1$, indicating agencies value preventing merger harm the same as investigating benign matters. Under this assumption, the thresholds are uniformly and substantially larger in magnitude.

The top half of Table 5 depicts the current rule (if available), the optimal rule, the value of the objective function at the optimum, and the variance of industry-wide prices in a

⁹We also experimented with a 10%/5% threshold and found that at those values no threshold existed for most of the measures.

Table 5: Optimal thresholds,by model

Model	Indicia	Current	Optimum	Value	Variance
Type I/Type II	Firm Count	5	6	1.7	17
	Party Share	30	32	1.8	23
	HHI	2,500	2,541	1.8	9.3
	HHI Change	200	463	1.9	10
	UPP		4.6	1.9	4.3
	CMCR		8.5	1.9	1.8
Agency	Firm Count	5	5	402,834	42
	Party Share	30	39	406,816	23
	HHI	2,500	3,059	404,654	23
	HHI Change	200	618	407,446	62
	UPP		6.7	408,281	9.5
	CMCR		16	408,512	7

neighborhood around the optimal decision rules when $\frac{V_H}{V_S} = 10$. Here, note that while UPP is the decision rule that maximizes the Type I/Type II decision rule, the other decision rules have thresholds that are quite close to the UPP. Further, note that the variance of industry-wide prices changes around the optimum varies across the indices, with Firm Count having the largest variance and CMCR the lowest.

4.2 Antitrust Agency Cost-Benefit Analysis

While a useful starting point for exploring optimal decision rules, the Type I/Type II error framework suffers from a two main drawbacks. First, this framework does not reflect any of the resource constraints that agencies typically operate under. While agencies do have substantial budgets, they are also tasked with evaluating a large and growing number of mergers each year. For example, the Antitrust Division’s budget increased by about 12% from \$148 million in 2008 to \$165 million in 2017, while the number of HSR filings the Division received during that period increased by about 19% from 1,726 filings in 2008 to 2,057 in 2017. These numbers suggest that the Division has fewer resources to spend per HSR filing.

Second, by focusing only on the percentage change in industry prices, the above framework ignores the fact that some mergers can have larger adverse effects because they occur in larger markets. For example, the above above framework treats a 1% industry-wide price increase in a \$1 billion dollar market as being benign while a 5% industry-wide price increase in a \$100 million dollar market as being anti-competitive, even though the first market yields

roughly \$10 million of harm per year while the second market yields roughly \$5 million. Here we develop screening thresholds that are more closely based on the amount of harm from a merger as well as the costs associated with investigating a merger.

Suppose in each year t , firms file N_t Hart-Scott-Rodino notices to merge with an enforcement agency. Further, suppose that when the agency is determining whether to open investigations, they weigh the expected benefit from opening an investigation into a merger against the costs associated with investigating the merger. We begin by identifying an optimal decision rule for determining whether to investigate a merger based on the change in the HHI.

Formally, suppose that candidate decision rule D_I based on index I , the expected net benefit to the agency is

$$E[NB_t|D_I] = \sum_{n=1}^{\infty} E[NB_t|D_I, N(t) = n] e^{-\lambda t} \frac{(\lambda t)^n}{n!} \quad (2)$$

$$E[NB_t|D_I, N(t) = n] = E\left[\sum_{j=1}^{d(n)} \left(H_j - \frac{j}{\kappa}\right) e^{-\rho S_j} | D_I\right] \quad (3)$$

where H_j is the harm from the j th merger, ρ is the discount rate, S_j is the arrival time of notice n , $\frac{j}{\kappa}$ is the incremental cost associated with investigating a merger, and the agency chooses to further investigate any merger with a change in index I greater than D_I . Since the number of Hart-Scott-Rodino filings varies from year to year, we suppose that $N(t)$ follows a Poisson distribution with rate λ . Assuming that the size of the merging parties is independent from when Hart-Scott-Rodino is filed, it can be shown that equation (3) reduces to¹⁰

$$E[NB_t|D_I] = \frac{2\kappa E[D_I H_n] - \lambda \Pr(I > D_I)(\lambda t \Pr(I > D_I) + 2)}{2\kappa\rho} \lambda(1 - e^{-\rho t}) \quad (4)$$

where $E[D_I H_n]$ is the expected amount of harm from a merger when the agency chooses to investigate it further under rule D_I . Hence, an optimal decision threshold D_I^* is one which maximizes (??) for a given indicia.

To calculate these thresholds, we first assume that the actual distribution of horizontal

¹⁰See Ross 2003.

mergers corresponds to the data generating process underlying our Monte Carlo simulations. Next, we calibrate κ by assuming that the 2017 average variable costs for the DOJ Antitrust Division are equal to the total number of HSRs received by the Division in 2017, divided by the Division’s expenditures on HSRs. This is calculated by multiplying the percentage of total matters initiated in 2017 that were HSRs by the Division’s 2017 budget. The value of κ is then backed out from the definition of average variable costs.

Figure 12 summarizes the results. The black curve depicts agency expected net benefit for a given threshold using (3). The green vertical line displays the current threshold, if available. The pink vertical line displays the threshold that maximizes (3). This figure depicts some interesting features of the optimization problem. First, the objective function achieves its highest values under UPP and CMCR, and its lowest value under Firm Count, suggesting that at least on average, UPP and CMCR outperform the other indicia. Second, CMCR has the “flattest” objective function, suggesting that it is the most robust to relatively small deviations away from the optimum.¹¹

Both of the qualitative points are more clearly presented in the bottom half of Table 5, which depicts the current rule (if available), the optimal rule, the value of the objective function at the optimum, and the variance of industry-wide prices in a neighborhood around the optimal decision rules when $\frac{V_H}{V_S} = 10$. Here, note that while CMCR is the decision rule that maximizes the Agency decision rule, the other decision rules have thresholds that are quite close to the CMCR. In addition, note that the variance of industry-wide price changes varies across the indices, with Firm Count again having the largest variance and CMCR the lowest.

4.3 Model composition and thresholds

One drawback of the decision rules depicted in Figures 11 and 12 is that they assume that the underlying population of mergers follows a particular model distribution: 30% Cournot, 50% Bertrand, and 20% Auction. Here, we explore how changing the model composition can affect these thresholds.

Table 6 reports that under the simple model with $\frac{V_H}{V_S} = 10$, the decision rules for each indicia vary substantially as the model mix changes. For example, the Firm Count threshold varies from 7 firms when Cournot comprises at least 50% of mergers, to 5 firms when Cournot

¹¹This is likely due to its explicit accounting for higher order effects on prices.

Table 6: Thresholds with different model weights, $\frac{V_H}{V_S} = 10$

Cournot	Bertrand	Auction	Firm Count	Party Gap	Party Share	HHI	HHI Change	UPP	CMCR
0.9	0.05	0.05	7	1	22	1,920	231	2.2	3.5
0.7	0.1	0.2	7	1	26	1,967	286	2.5	3.8
0.5	0.3	0.2	7	1	28	2,190	332	3.5	5.7
0.5	0.1	0.4	7	1	29	2,223	374	3.1	4.8
0.3	0.5	0.2	6	1	32	2,465	454	4.6	8.5
0.27	0.55	0.18	6	1	32	2,542	463	4.6	8.5
0.3	0.3	0.4	6	1	32	2,542	463	4.3	8.5
0.3	0.1	0.6	6	1	32	2,562	464	4.4	8.4
0.1	0.7	0.2	5	1	40	2,943	634	6.4	14
0.1	0.5	0.4	5	1	40	3,013	657	7.5	17
0.1	0.3	0.6	5	1	40	3,103	666	8.1	18
0.1	0.1	0.8	5	1	41	3,464	733	8.6	20

comprises only 10% of mergers. Similarly, the Party Share threshold varies from 25% when Cournot comprises 90% of mergers, but is at least 40% when Cournot comprises only 10% of mergers. Interestingly, The HHI Change threshold only approaches the current level of 200 when Cournot comprises 90% of mergers.

Similar patterns occur in Table 7, which reports comparable results for the Agency model. For example, the Firm Count threshold varies from 7 firms when Cournot comprises 90% of mergers, to 4 firms when Cournot comprises only 10% of mergers. The Party Share threshold varies from 26% when Cournot comprises 90% of mergers, but is at least 47% when Cournot comprises only 10% of mergers. Interestingly, the post-merger HHI threshold only approaches the current threshold of 2,500 when Cournot comprises at least 50% of mergers, and the HHI Change threshold never comes close to the current level of 200. Taken together these observations suggest that the current agency thresholds are best suited to a population of mergers in which industries engaged in Cournot competition dominate.

5 Conclusion

This paper contributes to understanding the relationship between horizontal merger investigation screening measures and predicted industry price effects from standard merger simulation models. We find that across a range of common demand and supply specifications all the concentration measures and other indicia of competitive harm correctly predict when a merger is likely to have little anti-competitive effect (i.e small values of the indicia are

Table 7: Thresholds with different model weights

Cournot	Bertrand	Auction	Firm Count	Party Gap	Party Share	HHI	HHI Change	UPP	CMCR
0.9	0.05	0.05	7	1	26	2,205	290	2.6	4.1
0.7	0.1	0.2	7	1	30	2,275	383	3.1	5.4
0.5	0.3	0.2	6	1	32	2,599	476	4.5	8.3
0.5	0.1	0.4	6	1	32	2,565	476	4.7	8.4
0.3	0.1	0.6	5	1	38	3,059	595	8.2	20
0.3	0.5	0.2	5	1	38	2,972	595	6.4	14
0.3	0.3	0.4	5	1	38	2,961	608	7.1	18
0.27	0.55	0.18	5	1	39	3,059	618	6.7	16
0.1	0.5	0.4	4	1	47	3,848	935	11	25
0.1	0.3	0.6	4	1	47	3,897	939	12	32
0.1	0.7	0.2	4	1	46	3,709	939	9	22
0.1	0.1	0.8	4	1	49	3,967	995	13	38

associated with small price effects). However, while large indicia values are associated with anti-competitive effects, in general these indicia do a poor job of accurately predicting price effects when they are likely to be large.

In addition, we consider optimal decision rules for merger enforcement based on commonly used measures of concentration and compare them to current screening thresholds employed by US antitrust enforcement agencies. We find that in this framework the current thresholds suggest a significantly higher value is placed on investigating potentially anticompetitive mergers than on avoiding investigating mergers that are unlikely to raise competitive concerns. In addition, the current thresholds would be most accurate in a population with the majority of industries engaged in Cournot competition.

An important implication of our Monte Carlo experiments is that measures that include information on substitution to the outside good (UPP, CMCR) appear to perform better than those that do not. This reflects the importance of appropriately defined alternatives to the products at issue in a merger, which is an under-appreciated driver of merger simulation results. While optimal decision thresholds can be developed for any competitive screening measure, the resulting decision thresholds perform better (both on average and in terms of dispersion) using indicia that incorporate substitution to the outside good as well. Precision of screening measures around optima varies, with the lowest variance in predicted price effects from CMCR, strengthening its case for use as a screening measure.

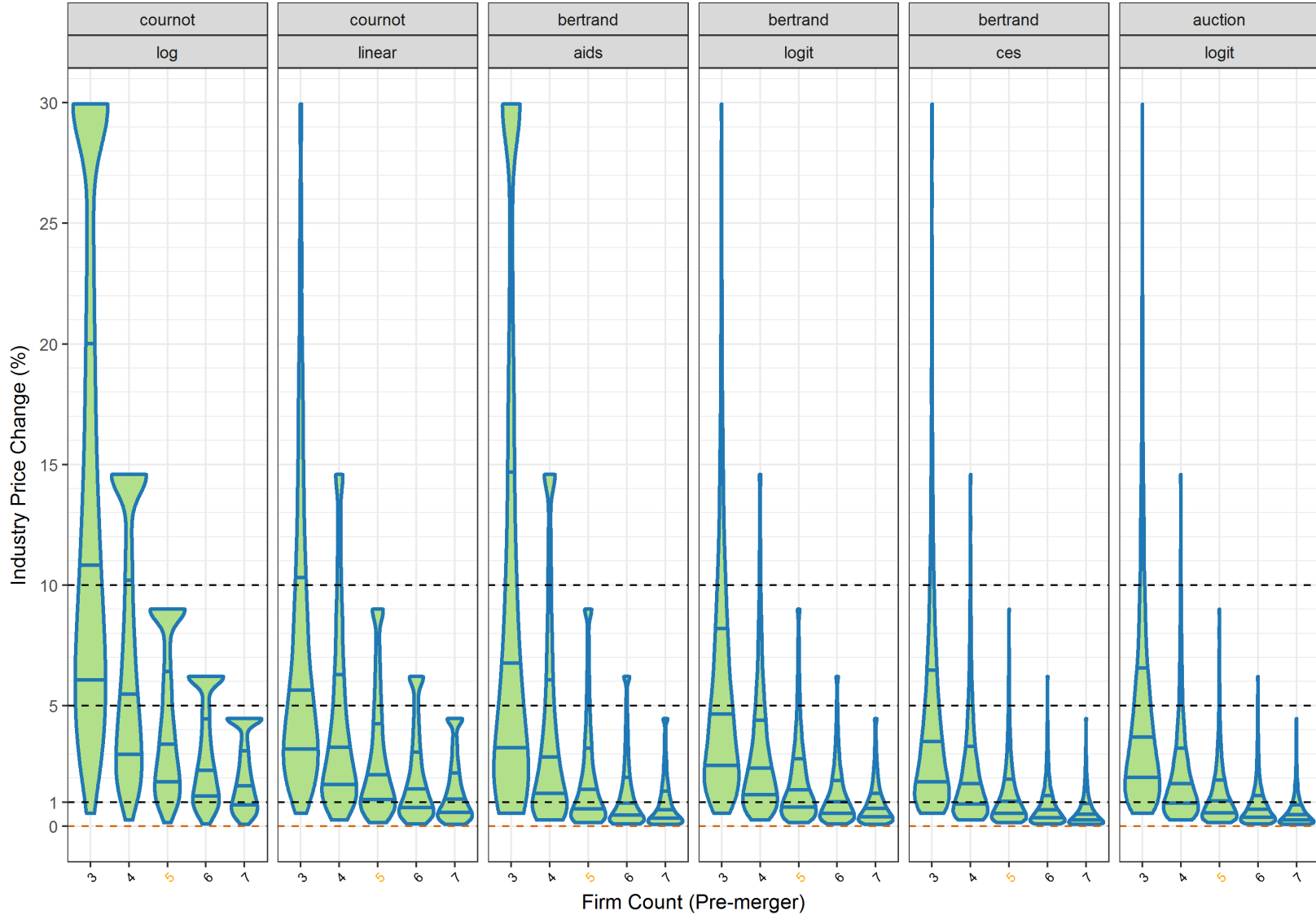


Figure 2 summarizes the relationship between number of pre-merger firms and industry price changes. Highlighted in orange is the 5-4 decision rule implied by the HHI thresholds in the 2010 HMG. Industry price changes are censored at the 5th and 95th percentile. Horizontal lines within each plot represent the 25th, 50th and 75th percentiles.

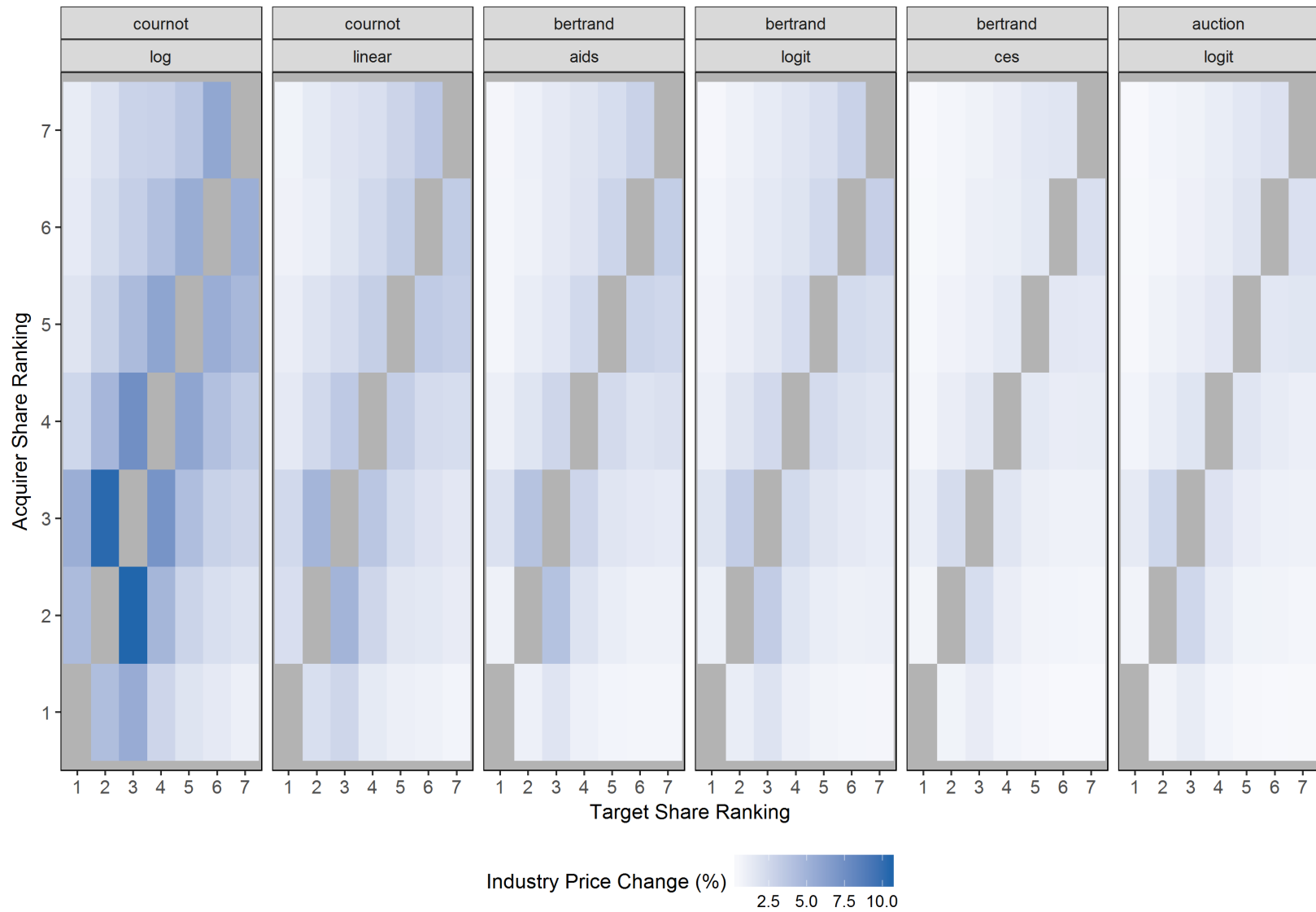


Figure 3 displays a heatmap summarizing the relationship between the share ranks of the merging parties and median industry price changes.

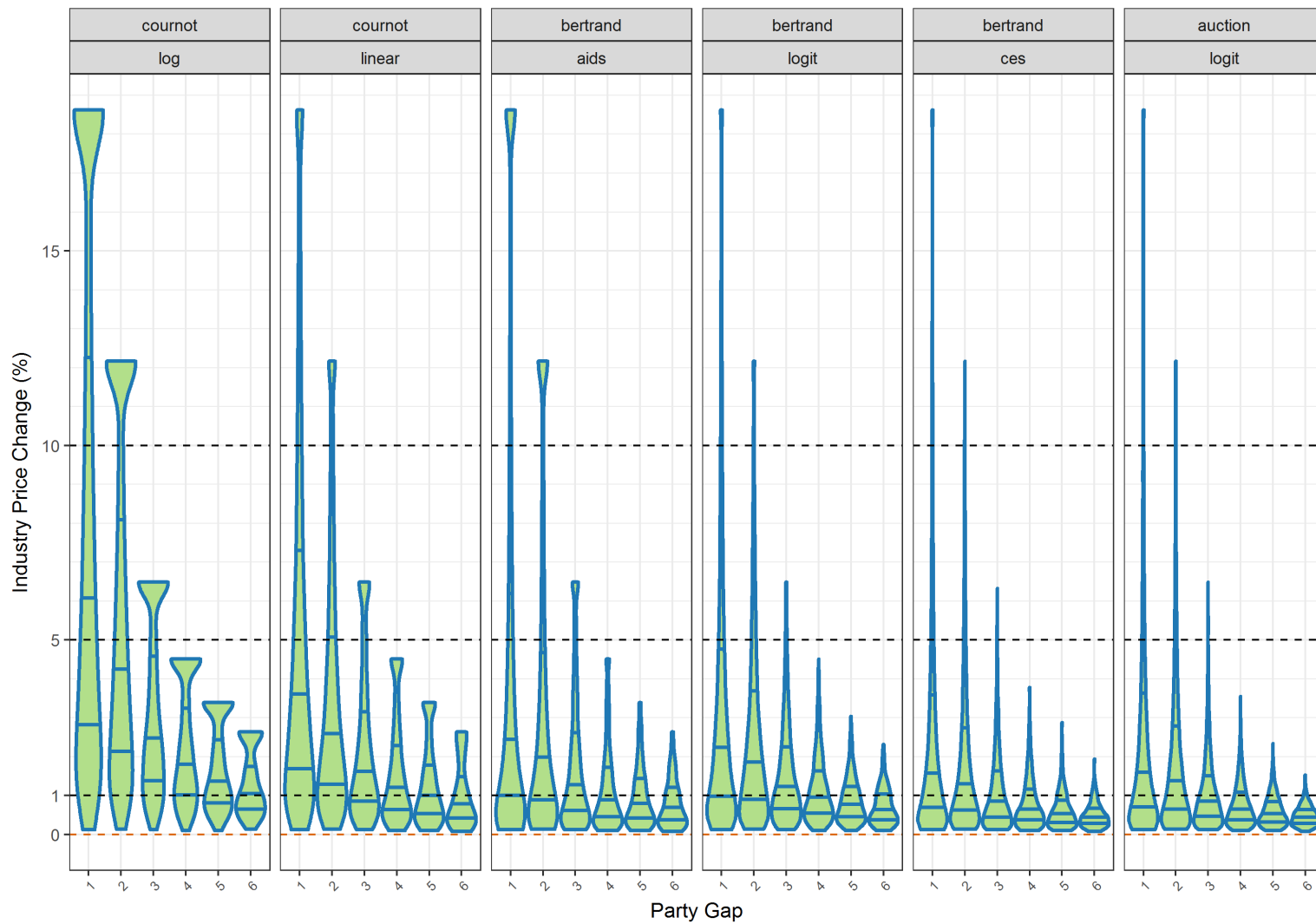


Figure 4 displays violin plots summarizing the relationship between the gap in party size rank and simulated industry price changes.

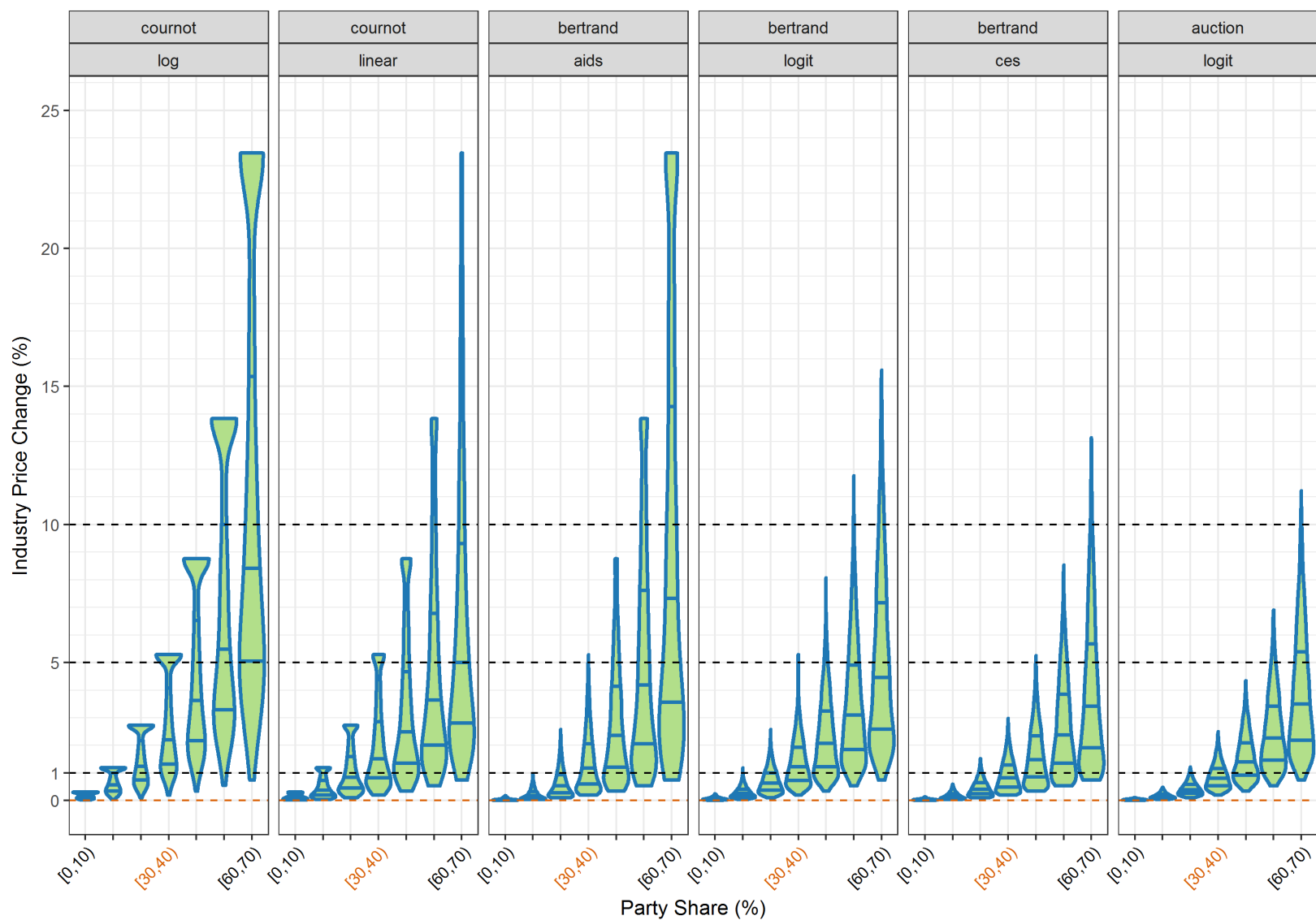


Figure 5 summarizes the relationship between the combined share of the merging parties and simulated industry price changes. Industry price changes have been censored at the 5th and 95th percentile. Horizontal lines in each plot depict the 25th, 50th and 75th percentiles. The bin containing combined shares of 30% and 35%, historically identified as thresholds for a presumptively anti-competitive merger, is colored orange.

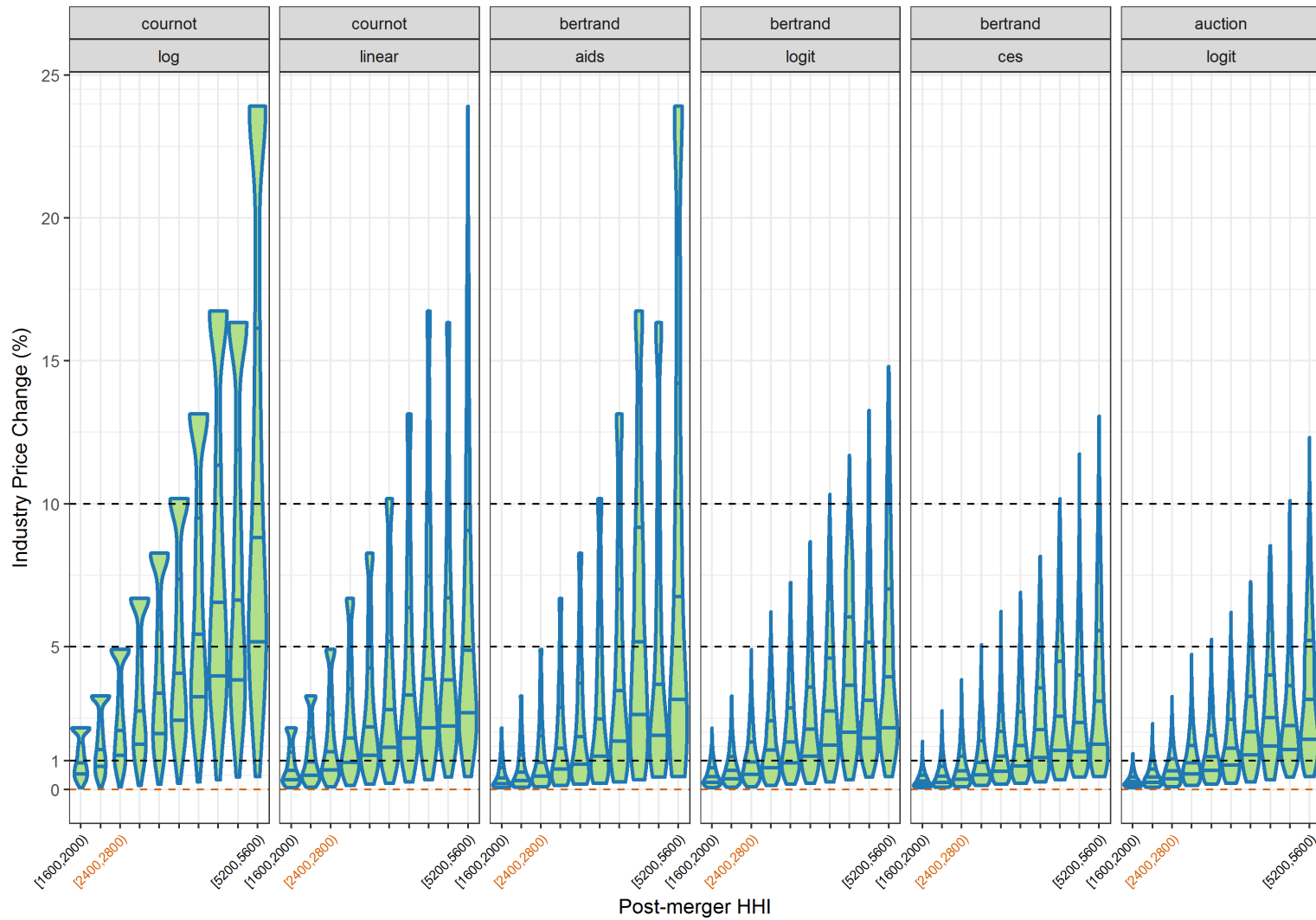


Figure 6 displays violin plots summarizing the relationship between the HHI and simulated industry price changes. Industry price changes have been censored at the 5th and 95th percentile, and the horizontal lines depicted in each plot represent the 25th, 50th and 75th percentiles. The bin containing the post-merger HHI of 2500, a component of establishing that a merger is presumptively anti-competitive under the HMG, is colored orange.

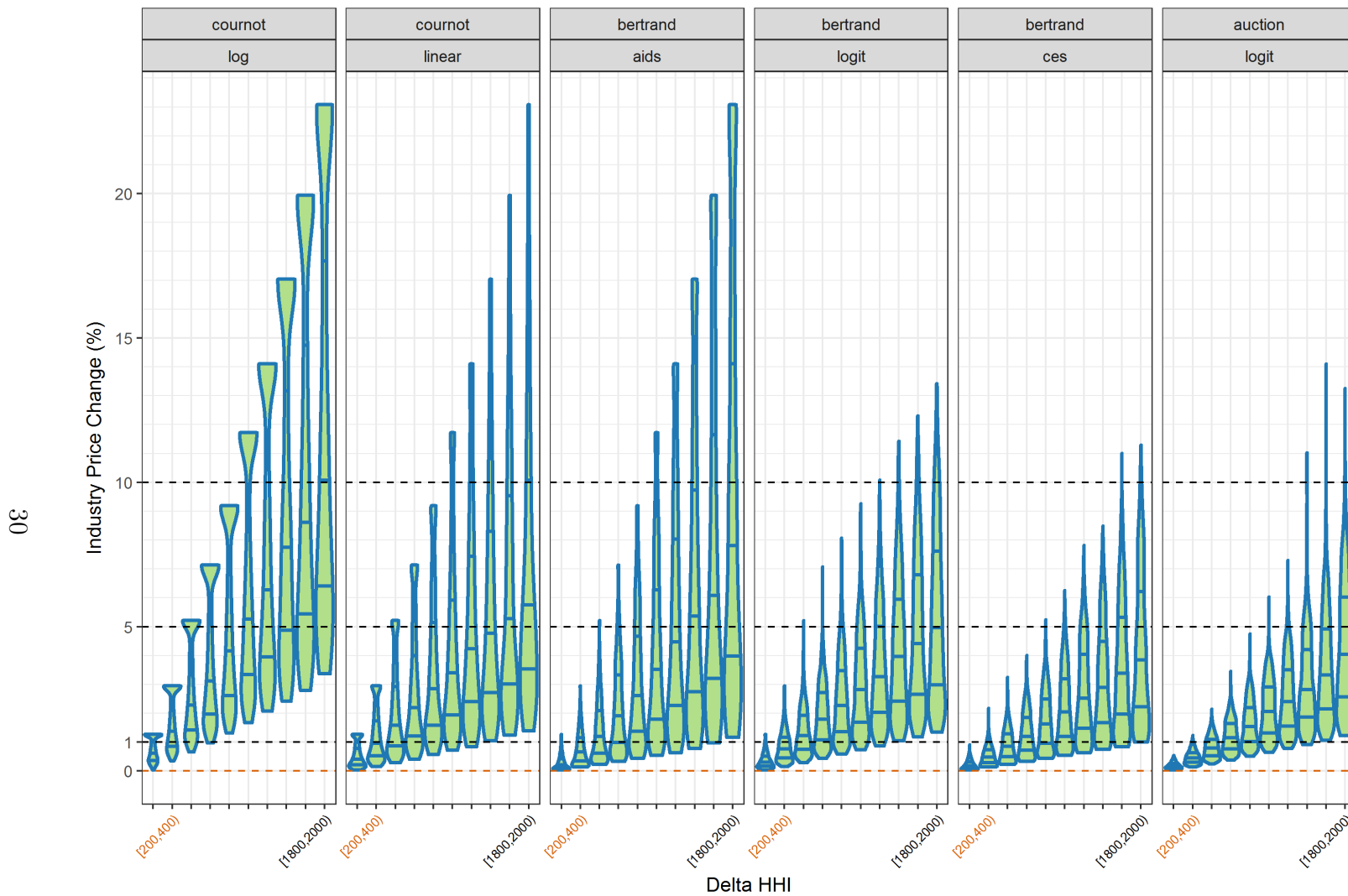


Figure 7 displays violin plots summarizing the relationship between changes in HHI and simulated industry price changes. Industry price changes have been censored at the 5th and 95th percentile, and the horizontal lines depicted in each plot represent the 25th, 50th and 75th percentiles. HHI change of 200, a component of establishing that a merger is presumptively anti-competitive under the HMG, is colored orange.

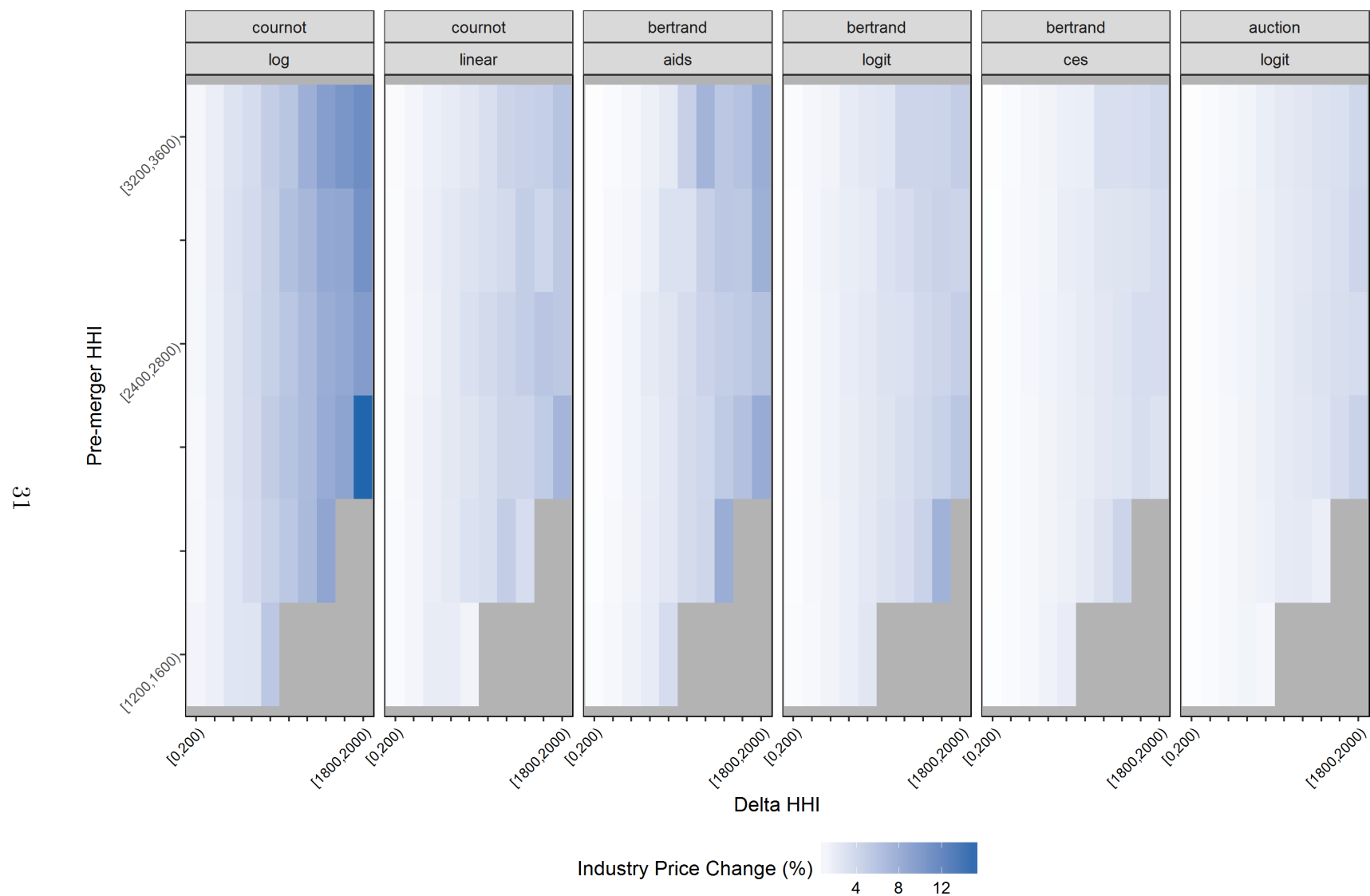


Figure 8 displays a heatmap summarizing the relationship between pre-merger HHI, changes in HHI and median industry price changes.

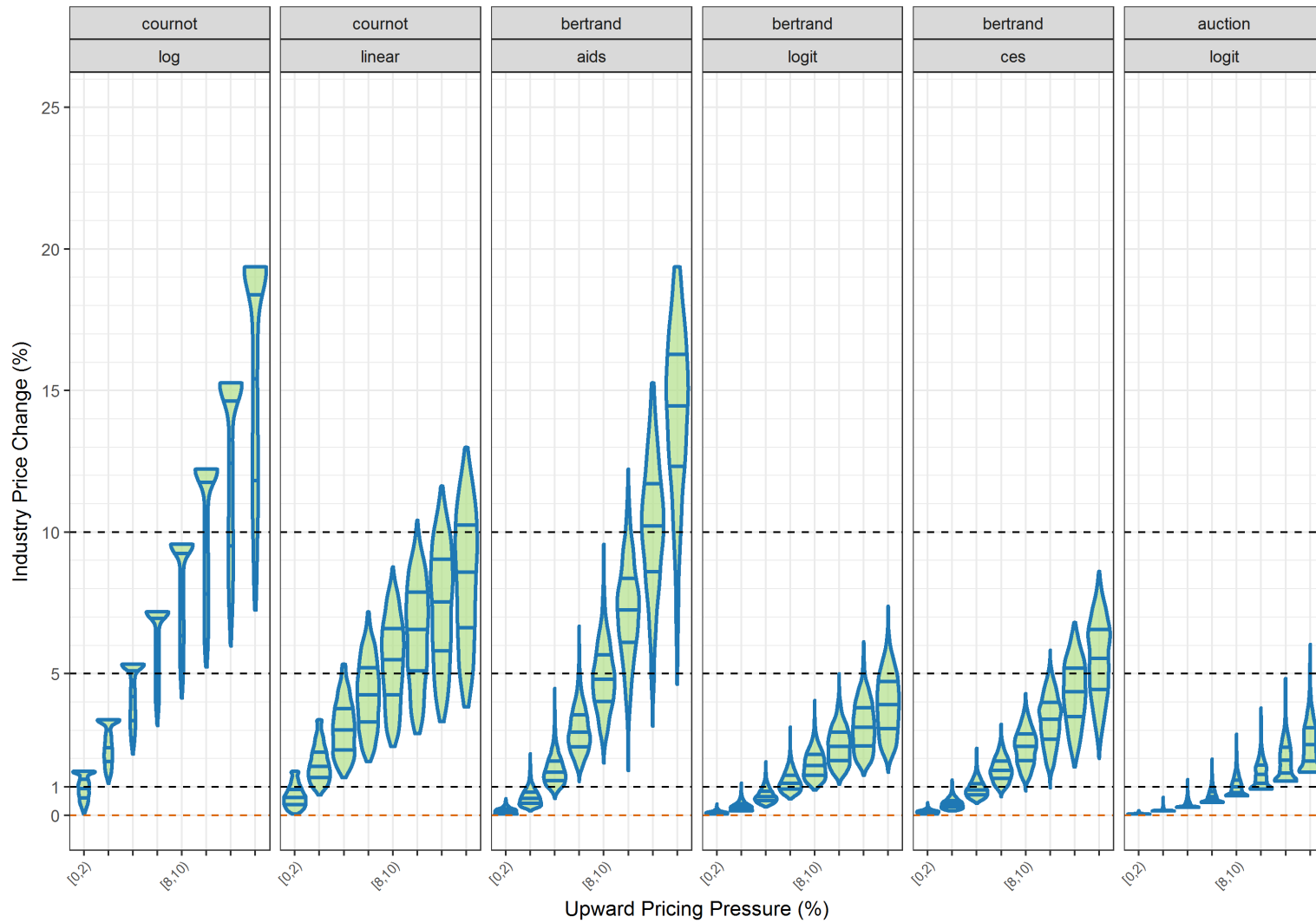


Figure 9 displays violin plots summarizing the relationship between the Upward Pricing Pressure index (UPP) and simulated industry price changes. Each panel summarizes the results from 50,000 Monte Carlo Simulations. Industry price changes have been censored at the 5th and 95th percentile, and the horizontal lines depicted in each plot represent the 25th, 50th and 75th percentiles. All simulations are run assuming that the outside good has a price of \$6 and \$2 marginal costs.

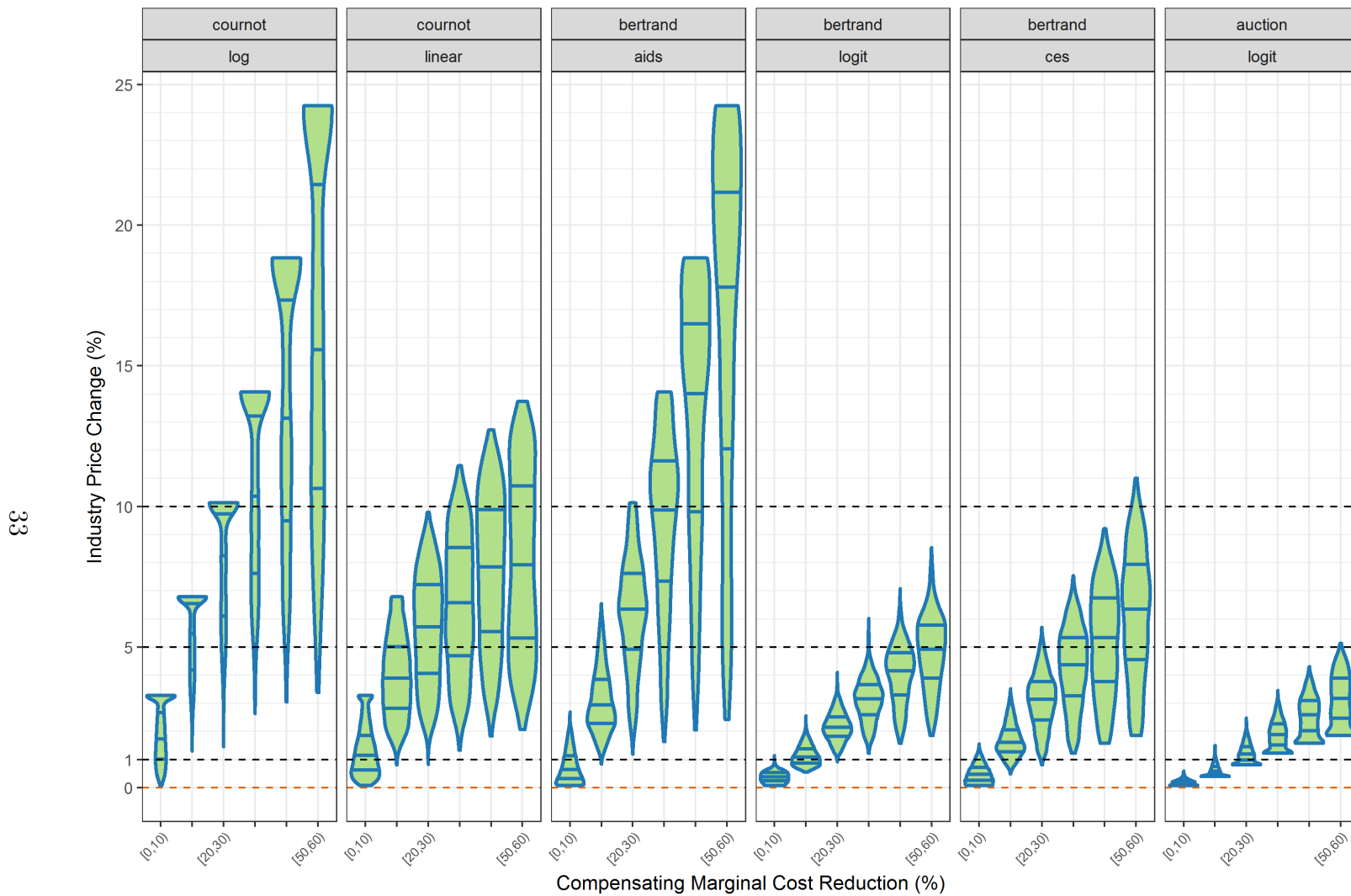


Figure 10 displays violin plots summarizing the relationship between the compensating marginal cost reduction (CMCR) and simulated industry price changes. Each panel summarizes the results from 50,000 Monte Carlo Simulations. Industry price changes have been censored at the 5th and 95th percentile, and the horizontal lines depicted in each plot represent the 25th, 50th and 75th percentiles. All simulations are run assuming that the outside good has a price of \$6 and \$2 marginal costs.

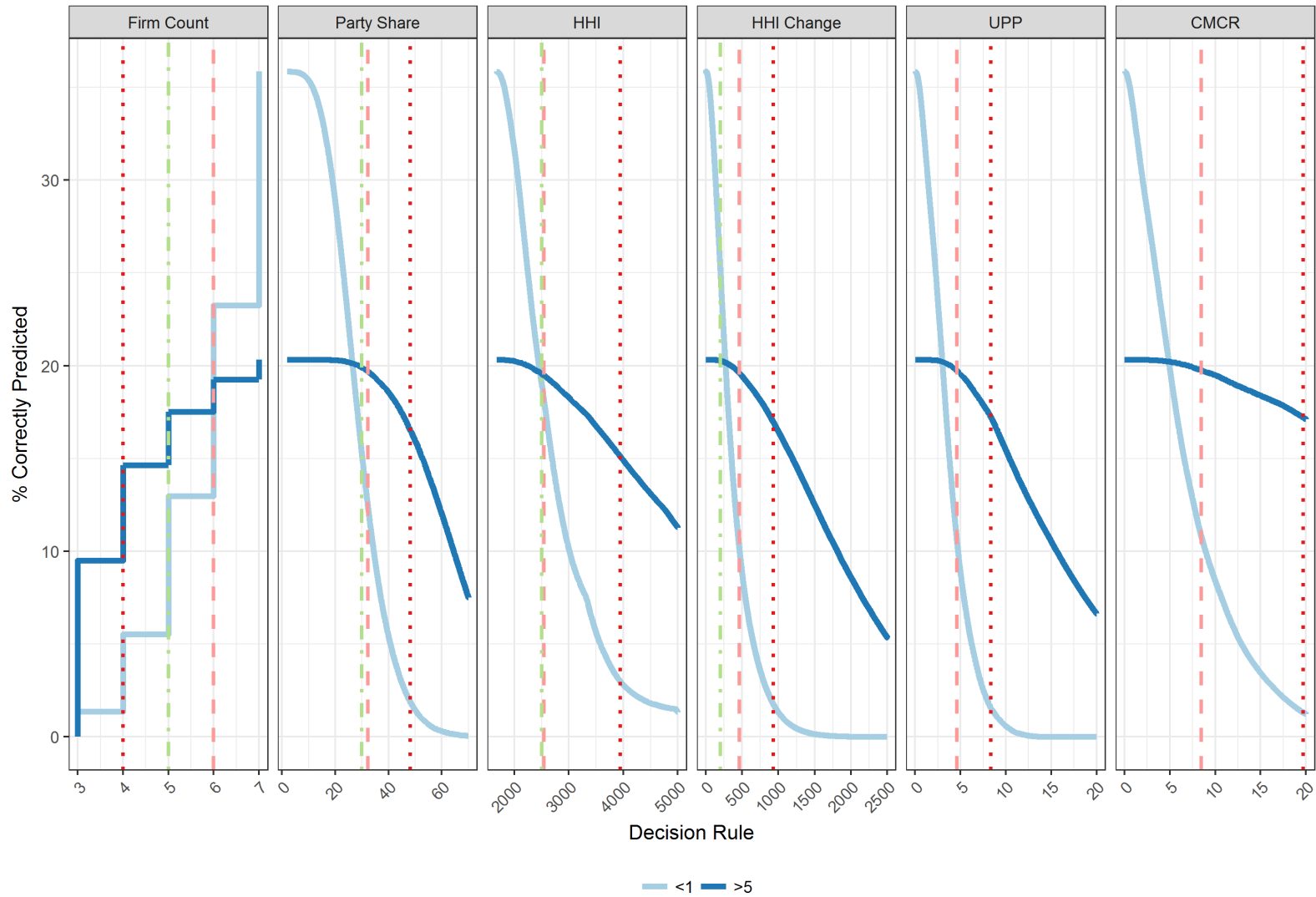


Figure 11 displays the *cumulative* percentage of markets exhibiting less than a 1% price increase (light blue) or greater than a 5% price increase (dark blue) for a given level of each indicia. Green (dot-dashed) vertical lines depict the currently used threshold where available. Pink (dashed) vertical lines depict the threshold that maximizes the gap between the light and dark blue lines, when the enforcement ratio equals 10. The red (dotted) vertical lines depict the threshold that maximizes the gap between the light and dark blue lines, when the enforcement ratio equals 1.

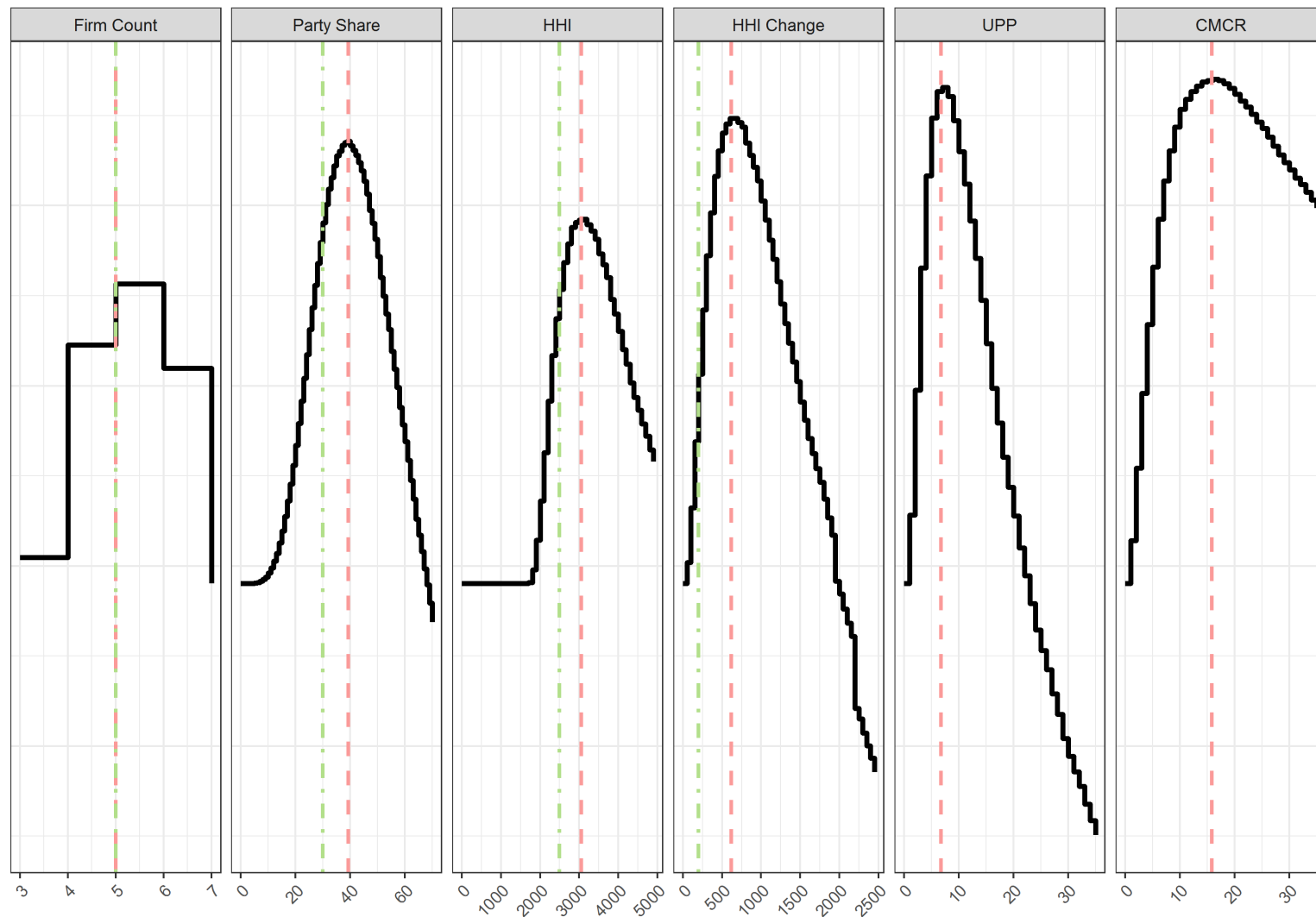


Figure 12 displays net benefit (harm averted less enforcement costs) to a hypothetical antitrust agency facing linear enforcement costs as function of each indicia. Green vertical lines depict the current threshold for each (where available). Pink vertical lines depict the threshold that maximize the agency's net benefit.

References

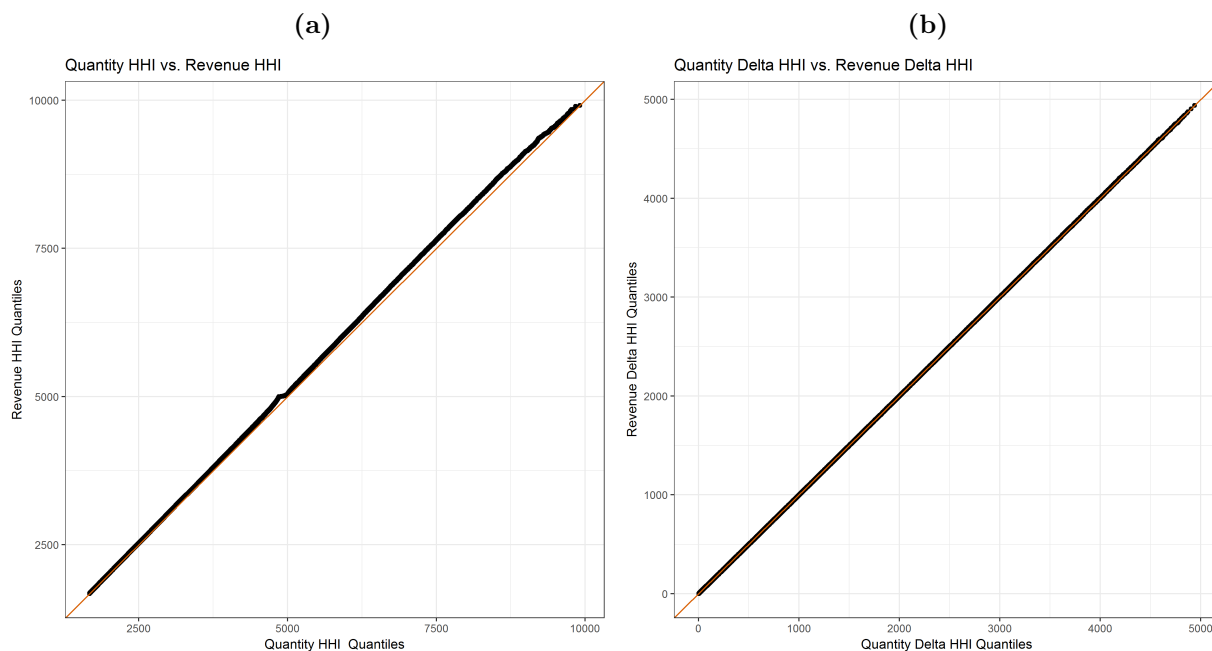
- [1] Bass, Frank et al. “What Happens When Demand is Estimated with a Misspecified Model”. In: *Journal of Industrial Economics* 56 (2008), pp. 809–829.
- [2] Severin Borenstein. “Hubs and High Fares: Dominance and Market Power in the U.S. Airline Industry”. In: *The RAND Journal of Economics* 20.3 (1989), pp. 344–365. ISSN: 07416261. URL: <http://www.jstor.org/stable/2555575>.
- [3] Lydia Cheung. “An Empirical Comparison Between the Upward Pricing Pressure Test and Merger Simulation in Differentiated Product Markets”. In: *Journal of Competition Law and Economics* 12.4 (2016), pp. 701–734.
- [4] Phillip Crooke et al. “The Effects of Assumed Demand Form on Simulated Post-Merger Equilibria”. In: *Review of Industrial Organization* 15 (2003), pp. 205–217.
- [5] Jerry A Hausman and J Gregory Sidak. “Evaluating market power using competitive benchmark prices instead of the Herfindahl-Hirschman index”. In: *Antitrust LJ* 74 (2007), p. 387.
- [6] Mark Manuszak and Charles Moul. “How Far for a Buck? Tax Differences and the Location of Retail Gasoline Activity in Southeast Chicagoland”. In: *The Review of Economics and Statistics* 91 (Nov. 2009), pp. 744–765. DOI: 10.1162/rest.91.4.744.
- [7] Mark D. Manuszak and Charles C. Moul. “Prices and Endogenous Market Structure in Office Supply Superstores”. In: *The Journal of Industrial Economics* 56.1 (2008), pp. 94–112. DOI: 10.1111/j.1467-6451.2008.00334.x. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-6451.2008.00334.x>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-6451.2008.00334.x>.
- [8] Nathan H. Miller et al. “Upward pricing pressure as a predictor of merger price effects”. In: *International Journal of Industrial Organization* 52.C (2017), pp. 216–247.
- [9] Joris Pinkse, Margaret E Slade, and Craig Brett. “Spatial price competition: a semi-parametric approach”. In: *Econometrica* 70.3 (2002), pp. 1111–1153.
- [10] Sheldon M. Ross. *Introduction to Probability Models*. Eighth. San Diego, CA, USA: Academic Press, 2003.

- [11] Richard Schmalensee. “Inter-industry studies of structure and performance”. In: *Handbook of Industrial Organization*. Ed. by R. Schmalensee and R. Willig. 1st ed. Vol. 2. Elsevier, 1989. Chap. 16, pp. 951–1009. URL: <https://EconPapers.repec.org/RePEc:eee:indchp:2-16>.
- [12] U.S. Department of Justice and Federal Trade Commission. *Horizontal Merger Guidelines*. 2010.
- [13] Matthew C. Weinberg and Daniel Hosken. “Evidence on the Accuracy of Merger Simulations”. In: *The Review of Economics and Statistics* 95.5 (2013), pp. 1584–1600.
- [14] Gregory J. Werden and Luke M. Froeb. “Simulation as an Alternative to Structural Merger Policy in Differentiated Products Industries”. In: *The Economics of the Antitrust Process*. Ed. by Malcolm B. Coate and Andrew N. Kleit. Boston, MA: Springer US, 1996, pp. 65–88.
- [15] Gregory J. Werden and Luke M. Froeb. “Unilateral Competitive Effects of Horizontal Mergers”. In: *Handbook of Antitrust Economics*. Ed. by Paolo Buccirossi. Cambridge, MA: MIT Press, 2006.

Appendix A Computing HHI: revenue vs. quantity shares

Does the measure of output for calculating market shares for use in HHI affect the ability to accurately predict competitive effects? Figure 13 displays the relationship between the quantiles of quantity- and revenue-based Herfindahl measures, depicted in a standard Q-Q plot. Panel 13a and 13b illustrates a nearly linear, monotonic relationship between both the levels and changes in the post-merger HHI calculated in terms of quantity or revenue shares, suggesting that the choice of measure may be based on convenience without altering the effect on this screening measure in this context.

Figure 13 Quantile-Quantile Plot of Revenue vs. Quantity based HHI



Appendix B Demand and Supply Specifications

B.1 Cournot

Suppose there are $N \in \{3, 4, 5, 6, 7\}$ firms playing a Cournot quantity setting game, where single-plant firms with quadratic cost technology $C_n(q_n) = \frac{q_n^2}{2k_n}$ simultaneously produce q_n units of a homogeneous product *once* to maximize their profits.

Mathematically, each firm n solves

$$\max_{q_n} Pq_n - C_n(q_n)$$

with demand equal to

$$P = \begin{cases} a - b^{-1}Q, & \text{if linear} \\ \exp(a)Q^{\frac{1}{\epsilon}}, & \text{if log-linear,} \end{cases}$$

where $Q = \sum_{j \in N} q_j$ denotes equilibrium output and $\epsilon = \frac{dQ}{dP} \frac{P}{Q}$ denotes the aggregate elasticity.

Define $m_n = -\frac{P - \frac{dC}{dq_n}}{P}$ as plant n 's proportional margin and $s_n = \frac{q_n}{Q}$ as plant n 's quantity share. Pre-merger, the first order condition (FOC) of each firm can be arranged to yield the Lerner condition $m_n = \frac{s_n}{\epsilon}$, which is an implicit function of all firms quantities. Under quadratic costs, it can be shown that this system of N equations has a unique Nash Equilibrium in quantities and that this equilibrium has a closed-form solution.

Moreover, the Lerner condition may be rearranged to yield an equilibrium relationship between industry average margins, HHI, and market elasticities. Specifically, taking the share-weighted average of all N firms' Lerner conditions yields

$$\begin{aligned} \sum_{n \in N} s_n m_n &= -\frac{\sum_{n \in N} s_n^2}{\epsilon} \\ &= \frac{HHI}{\epsilon} \end{aligned}$$

To calibrate the model parameters, we first assume that market size M is known and draw

N shares from a Dirichlet distribution. Second, we assume that in the pre-merger state, the equilibrium price is \$6 and there is an outside Firm 0 playing the Cournot game earning an equilibrium margin of \$2 with quantity share s_0 drawn from a uniform distribution between 0.1 and 0.7. We use Firm 0's Lerner condition to solve for the aggregate elasticity, and then the assumed price and aggregate output to solve for the remaining demand parameters. Finally, we again use the Lerner condition, quantities and price to solve for the cost parameters k_n ¹².

Next, suppose that Firm 1 acquires Firm 2. We simulate the effect of the merger by assuming that Firm 1 sets output on plants 1 and 2 jointly. Mathematically, Firm 1 solves

$$\max_{q_1, q_2} P(q_1 + q_2) - C_1(q_1) - C_2(q_2)$$

Taking the FOC and rearranging yields

$$m_1 = m_2 = \frac{s_1 + s_2}{\epsilon}$$

The functional form of the non-merging parties' Lerner condition does not change as a result of the merger. The outside product's output is assumed not to adjust post-merger.

B.2 Bertrand

Suppose there are $N \in \{3, 4, 5, 6, 7\}$ firms playing a Bertrand price setting game, where single-product firms with constant marginal cost technology $C_n(q_n) = c_n q_n$ simultaneously set their prices p_n *once* to maximize their profits.

Mathematically, each firm $n \in N$ solves

$$\max_{p_n} (p_n - c_n)q_n$$

Like Cournot, the pre-merger first order condition (FOC) of each firm can be arranged to yield the Lerner condition $m_n = \frac{1}{\epsilon_{nn}}$, which is an implicit function of all firms prices. In general, this system of equations has no closed form solution, but a Nash equilibrium in prices may be solved for numerically.

¹²From time to time, this procedure yielded plant margins that were greater than 1. When this occurred, we simply discarded the market and sampled another one.

Moreover, it has been shown that the magnitude of the price effects in the Bertrand model depends critically on the second order properties of the demand curve. These properties are captured by the own-price elasticities $\epsilon_{nn} = -\frac{\partial q_n}{\partial p_n} \frac{p_n}{q_n}$. Specifically, we will consider the properties of three popular demand systems: Logit, CES, and AIDS

B.2.1 Logit Demand

Logit demand is based on a discrete choice model that assumes that each consumer is willing to purchase at most a single unit of one product from the N products available in the market. The assumptions underlying Logit demand imply that the probability that a consumer purchases product $n \in N$ is given by

$$s_n = \frac{\exp(V_n)}{\sum_{k \in N} \exp(V_k)}$$

where s_n is product n 's *quantity* share and V_n is the (average) indirect utility that a consumer receives from purchasing product n . We assume that V_n takes on the following form

$$V_n = \delta_n + \alpha p_n, \quad \alpha < 0$$

The Logit demand system yields the following own- and cross-price elasticities:

$$\begin{aligned} \epsilon_{nn} &= \alpha(1 - s_n)p_n \\ \epsilon_{nj} &= -\alpha s_j p_j \end{aligned}$$

B.2.2 CES Demand

Like the Logit, CES demand is based on a discrete choice model. However, CES differs from the Logit model in that under CES consumers do not purchase a single unit of a product but instead spend a fixed proportion of their budget on one of the n products available in

the market.¹³

The assumptions underlying CES demand imply that the probability that a consumer purchases product $n \in N$ is given by

$$r_n = \frac{V_n}{\sum_{k \in N} V_k} \quad \text{for all } n \in N$$

where r_n is product n 's *revenue* share and V_n is the (average) indirect utility that a consumer receives from purchasing product n . We assume that V_n takes on the following form

$$V_n = \delta_n p_n^{1-\gamma}, \quad \gamma > 1$$

The CES demand system yields the following own- and cross-price elasticities:

$$\begin{aligned} \epsilon_{nn} &= -\gamma + (\gamma - 1)r_n \\ \epsilon_{nj} &= (\gamma - 1)r_j \end{aligned}$$

Functional form differences aside, one important difference between the CES and Logit demand systems is that the Logit model's choice probabilities are expressed in terms of *quantity* shares, while the CES model's choice probabilities are expressed in terms of *revenue* shares.

B.2.3 AIDS

AIDS without income effects assumes that the demand for each product $n \in \{N, 0\}$ in the market is given by

¹³Formally, each consumer chooses the product $n \in N$ that yields the maximum utility $U_n = \ln(\delta_n q_n) + \alpha \ln(q_0) + \epsilon_i$, subject to the budget constraint $y = p_n q_n + q_0$. Here, q_n is the amount of product n consumed by a consumer, δ_n is a measure of product n 's quality, q_0 is the amount of the numeraire, y is consumer income, and ϵ_n are random variables independently and identically distributed according to the Type I Extreme Value distribution.

$$r_n = \alpha_n + \sum_{n \in \{N, 0\}} \beta_{nj} \log(p_j) \text{ for all } n \in \{N, 0\}, \quad \beta_{nn} < 0$$

and total expenditure x is given by

$$\log(x) = \eta + \sum_{j \in N} \alpha_j \log(p_j) + \sum_{k \in N} \sum_{j \in N} \beta_{kj} \log(p_k) \log(p_j)$$

The AIDS model yields the following own- and cross-price elasticities:

$$\begin{aligned} \epsilon_{nn} &= -1 + \frac{\beta_{nn}}{r_n} + r_n, \quad \epsilon_{nn} < 0 \\ \epsilon_{nj} &= \frac{\beta_{nj}}{r_n} + r_j, \quad \epsilon_{nj} \geq 0 \end{aligned}$$

This version of the AIDS model assumes that $\beta_{nj} = \beta_{jj}$, satisfies homogeneity of degree zero in prices, and that diversion occurs according to revenue share $d_{jn} = -\frac{r_n}{1-r_j} = -\frac{\beta_{nj}}{\beta_{jj}}$.